Solve Lambda Field Equations

Case 5D_1A : 4D spacetime (t,r,θ,ϕ) + one curled axis (ψ)
1: Curled geometry: Circle
A: Curled metric - a[t]^2 $\mathcal{D}$[r] p^2 dψ^2
3-space metric: -B(t,r) dr^2 – $\mathbb{C}$(t,r) r^2 dθ^2 – $\mathbb{C}$(t,r) r^2 sin$^2\theta$ dϕ^2
Time metric: A(t,r) dt^2

Line Element

Out[ ]=

$$ds^2 = A[t, r]\ dt^2 - B[t, r]\ dr^2 - r^2\,\mathbb{C}[t, r]\ d\theta^2 - r^2\,\mathrm{Sin}[\theta]^2\,\mathbb{C}[t, r]\ d\phi^2 - p^2\,a[t]^2\,\mathcal{D}[r]\ d\psi^2$$

Metric Tensor Matrices

Out[ ]=

$$g:\ (g_{\mu\nu}) = \begin{pmatrix} \frac{A}{c^2} & 0 & 0 & 0 & 0 \\ 0 & -B & 0 & 0 & 0 \\ 0 & 0 & -r^2\,\mathbb{C} & 0 & 0 \\ 0 & 0 & 0 & -r^2\,\mathbb{C}\,\mathrm{Sin}[\theta]^2 & 0 \\ 0 & 0 & 0 & 0 & -a^2\,p^2\,\mathcal{D} \end{pmatrix}$$

$$G:\ (g^{\mu\nu}) = \begin{pmatrix} \frac{c^2}{A} & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{B} & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{r^2\,\mathbb{C}} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\mathrm{Csc}[\theta]^2}{r^2\,\mathbb{C}} & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{a^2\,p^2\,\mathcal{D}} \end{pmatrix}$$

Case 5D_1A: 5D = 4D spacetime (t,r,θ,ϕ) + one curled axis (ψ)
$g_{00}=\frac{A(t,r)}{c^2}$, $g_{11}=-B(t,r)$, $g_{22}=-\mathbb{C}(t,r)r^2$, $g_{33}=-\mathbb{C}(t,r)r^2\sin^2\theta$
$g_{44}=-a(t)^2\mathcal{D}(r)p^2$, a(t) & $\mathcal{D}$(r) not constant

The Lambda field equations can be represented as $\frac{R_{\mu\nu}}{g_{\mu\nu}} = \Lambda$:

$$R_{00}:\ \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{c^2\,\dot{A}\,\dot{B}}{4\,A^2\,B} - \frac{c^2\,\dot{B}^2}{4\,A\,B^2} + \frac{c^2\,\ddot{B}}{2\,A\,B} - \frac{c^2\,\dot{A}\,\dot{\mathbb{C}}}{2\,A^2\,\mathbb{C}} - \frac{c^2\,\dot{\mathbb{C}}^2}{2\,A\,\mathbb{C}^2} + \frac{c^2\,\ddot{\mathbb{C}}}{A\,\mathbb{C}} - \frac{A'}{A\,B\,r} + \frac{(A')^2}{4\,A^2\,B} + \frac{A'\,B'}{4\,A\,B^2} - \frac{A'\,\mathcal{D}'}{4\,A\,B\,\mathcal{D}} - \frac{A'\,\mathbb{C}'}{2\,A\,B\,\mathbb{C}} - \frac{A''}{2\,A\,B} = \Lambda$$

$$R_{01}:\ -\frac{\dot{B}}{B\,r} + \frac{\dot{\mathbb{C}}}{r\,\mathbb{C}} - \frac{\dot{a}\,A'}{2\,a\,A} - \frac{\dot{\mathbb{C}}\,A'}{2\,A\,\mathbb{C}} + \frac{\dot{a}\,\mathcal{D}'}{2\,a\,\mathcal{D}} - \frac{\dot{B}\,\mathcal{D}'}{4\,B\,\mathcal{D}} - \frac{\dot{B}\,\mathbb{C}'}{2\,B\,\mathbb{C}} - \frac{\dot{\mathbb{C}}\,\mathbb{C}'}{2\,\mathbb{C}^2} + \frac{\dot{\mathbb{C}}'}{\mathbb{C}} = 0$$

$$R_{11}: \quad \frac{c^2\,\dot{a}\,\dot{B}}{2\,a\,A\,B} - \frac{c^2\,\dot{A}\,\dot{B}}{4\,A^2\,B} - \frac{c^2\,\dot{B}^2}{4\,A\,B^2} + \frac{c^2\,\ddot{B}}{2\,A\,B} + \frac{c^2\,\dot{B}\,\dot{\mathbb{C}}}{2\,A\,B\,\mathbb{C}} + \frac{(A')^2}{4\,A^2\,B} + \frac{B'}{B^2\,r} + \frac{A'\,B'}{4\,A\,B^2} + \frac{B'\,\mathbb{D}'}{4\,B^2\,\mathbb{D}} + \frac{(\mathbb{D}')^2}{4\,B\,\mathbb{D}^2} - \frac{2\,\mathbb{C}'}{B\,r\,\mathbb{C}} + \frac{B'\,\mathbb{C}'}{2\,B^2\,\mathbb{C}} + \frac{(\mathbb{C}')^2}{2\,B\,\mathbb{C}^2} - \frac{A''}{2\,A\,B} - \frac{\mathbb{D}''}{2\,B\,\mathbb{D}} - \frac{\mathbb{C}''}{B\,\mathbb{C}} = \Lambda$$

$$R_{22}: \quad -\frac{1}{B\,r^2} + \frac{1}{r^2\,\mathbb{C}} + \frac{c^2\,\dot{a}\,\dot{\mathbb{C}}}{2\,a\,A\,\mathbb{C}} - \frac{c^2\,\dot{A}\,\dot{\mathbb{C}}}{4\,A^2\,\mathbb{C}} + \frac{c^2\,\dot{B}\,\dot{\mathbb{C}}}{4\,A\,B\,\mathbb{C}} + \frac{c^2\,\ddot{\mathbb{C}}}{2\,A\,\mathbb{C}} - \frac{A'}{2\,A\,B\,r} + \frac{B'}{2\,B^2\,r} - \frac{\mathbb{D}'}{2\,B\,r\,\mathbb{D}} - \frac{2\,\mathbb{C}'}{B\,r\,\mathbb{C}} - \frac{A'\,\mathbb{C}'}{4\,A\,B\,\mathbb{C}} + \frac{B'\,\mathbb{C}'}{4\,B^2\,\mathbb{C}} - \frac{\mathbb{D}'\,\mathbb{C}'}{4\,B\,\mathbb{D}\,\mathbb{C}} - \frac{\mathbb{C}''}{2\,B\,\mathbb{C}} = \Lambda$$

$$R_{44}: \quad \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} + \frac{c^2\,\dot{a}\,\dot{B}}{2\,a\,A\,B} + \frac{c^2\,\dot{a}\,\dot{\mathbb{C}}}{a\,A\,\mathbb{C}} - \frac{\mathbb{D}'}{B\,r\,\mathbb{D}} - \frac{A'\,\mathbb{D}'}{4\,A\,B\,\mathbb{D}} + \frac{B'\,\mathbb{D}'}{4\,B^2\,\mathbb{D}} + \frac{(\mathbb{D}')^2}{4\,B\,\mathbb{D}^2} - \frac{\mathbb{D}'\,\mathbb{C}'}{2\,B\,\mathbb{D}\,\mathbb{C}} - \frac{\mathbb{D}''}{2\,B\,\mathbb{D}} = \Lambda$$

We observed in Notebook 5D_1 that R_{33} is redundant
and, so, R_{33} is omitted here and in all of the 5D notebooks.

Beyond that, these equations contain all of the terms
found in Case 4D plus much more, so we already have discovered these
equations are too complex to solve without additional assumptions.

a: $\mathbb{B}(t,r)=a(t)^2\mathbb{B}(r)$, $\mathbb{C}(t,r)=a(t)^2\mathbb{C}(r)$

The field equations become

$$R_{00}: \quad \frac{4\,c^2\,\ddot{a}}{a\,A} - \frac{2\,c^2\,\dot{a}\,\dot{A}}{a\,A^2} - \frac{A'}{a^2\,A\,r\,\mathbb{B}} + \frac{(A')^2}{4\,a^2\,A^2\,\mathbb{B}} - \frac{A'\,\mathbb{C}'}{2\,a^2\,A\,\mathbb{C}\,\mathbb{B}} - \frac{A'\,\mathbb{D}'}{4\,a^2\,A\,\mathbb{D}\,\mathbb{B}} + \frac{A'\,\mathbb{B}'}{4\,a^2\,A\,\mathbb{B}^2} - \frac{A''}{2\,a^2\,A\,\mathbb{B}} = \Lambda$$

$$R_{01}: \quad -\frac{3\,\dot{a}\,A'}{2\,a\,A} = 0$$

$$R_{11}: \quad \frac{3\,c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} + \frac{(A')^2}{4\,a^2\,A^2\,\mathbb{B}} - \frac{2\,\mathbb{C}'}{a^2\,r\,\mathbb{C}\,\mathbb{B}} + \frac{(\mathbb{C}')^2}{2\,a^2\,\mathbb{C}^2\,\mathbb{B}} + \frac{(\mathbb{D}')^2}{4\,a^2\,\mathbb{D}^2\,\mathbb{B}} + \frac{\mathbb{B}'}{a^2\,r\,\mathbb{B}^2} + \frac{A'\,\mathbb{B}'}{4\,a^2\,A\,\mathbb{B}^2} + \frac{\mathbb{C}'\,\mathbb{B}'}{2\,a^2\,\mathbb{C}\,\mathbb{B}^2} + \frac{\mathbb{D}'\,\mathbb{B}'}{4\,a^2\,\mathbb{D}\,\mathbb{B}^2} - \frac{A''}{2\,a^2\,A\,\mathbb{B}} - \frac{\mathbb{C}''}{a^2\,\mathbb{C}\,\mathbb{B}} - \frac{\mathbb{D}''}{2\,a^2\,\mathbb{D}\,\mathbb{B}} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2\,\mathbb{C}} - \frac{1}{a^2\,r^2\,\mathbb{B}} + \frac{3\,c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{A'}{2\,a^2\,A\,r\,\mathbb{B}} - \frac{2\,\mathbb{C}'}{a^2\,r\,\mathbb{C}\,\mathbb{B}} - \frac{A'\,\mathbb{C}'}{4\,a^2\,A\,\mathbb{C}\,\mathbb{B}} - \frac{\mathbb{D}'}{2\,a^2\,r\,\mathbb{D}\,\mathbb{B}} - \frac{\mathbb{C}'\,\mathbb{D}'}{4\,a^2\,\mathbb{C}\,\mathbb{D}\,\mathbb{B}} + \frac{\mathbb{B}'}{2\,a^2\,r\,\mathbb{B}^2} + \frac{\mathbb{C}'\,\mathbb{B}'}{4\,a^2\,\mathbb{C}\,\mathbb{B}^2} - \frac{\mathbb{C}''}{2\,a^2\,\mathbb{C}\,\mathbb{B}} = \Lambda$$

$$R_{44}: \quad \frac{3\,c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{\mathbb{D}'}{a^2\,r\,\mathbb{D}\,\mathbb{B}} - \frac{A'\,\mathbb{D}'}{4\,a^2\,A\,\mathbb{D}\,\mathbb{B}} - \frac{\mathbb{C}'\,\mathbb{D}'}{2\,a^2\,\mathbb{C}\,\mathbb{D}\,\mathbb{B}} + \frac{(\mathbb{D}')^2}{4\,a^2\,\mathbb{D}^2\,\mathbb{B}} + \frac{\mathbb{D}'\,\mathbb{B}'}{4\,a^2\,\mathbb{D}\,\mathbb{B}^2} - \frac{\mathbb{D}''}{2\,a^2\,\mathbb{D}\,\mathbb{B}} = \Lambda$$

Since $a(t)$ is not allowed to be a constant in Case a, from R_{01} we get that

$$A(t,r) = \mathcal{A}(t)$$

R_{01} is eliminated, and the remaining equations become

$$R_{00}: \quad \frac{4\,c^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{2\,c^2\,\dot{a}\,\dot{\mathcal{A}}}{a\,\mathcal{A}^2} = \Lambda$$

$$R_{11}: \quad \frac{3 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{2 C'}{a^2 r C B} + \frac{(C')^2}{2 a^2 C^2 B} + \frac{(D')^2}{4 a^2 D^2 B} + \frac{B'}{a^2 r B^2} + \frac{C' B'}{2 a^2 C B^2} + \frac{D' B'}{4 a^2 D B^2} - \frac{C''}{a^2 C B} - \frac{D''}{2 a^2 D B} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2 C} - \frac{1}{a^2 r^2 B} + \frac{3 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{2 C'}{a^2 r C B} - \frac{D'}{2 a^2 r D B} - \frac{C' D'}{4 a^2 C D B} + \frac{B'}{2 a^2 r B^2} + \frac{C' B'}{4 a^2 C B^2} - \frac{C''}{2 a^2 C B} = \Lambda$$

$$R_{44}: \quad \frac{3 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{D'}{a^2 r D B} - \frac{C' D'}{2 a^2 C D B} + \frac{(D')^2}{4 a^2 D^2 B} + \frac{D' B'}{4 a^2 D B^2} - \frac{D''}{2 a^2 D B} = \Lambda$$

We can solve R_{00} for $\mathcal{A}(t)$:

$$\mathcal{A}(t) = \frac{4 c^2 \dot{a}^2}{4 c^2 k + a^2 \Lambda}$$ where k is a constant of integration.

Plugging this gives us 3 equations in the 3 unknowns $B(r)$, $C(r)$, and $D(r)$:

$$R_{11}: \quad \frac{3 c^2 k}{a^2} + \Lambda - \frac{2 C'}{a^2 r C B} + \frac{(C')^2}{2 a^2 C^2 B} + \frac{(D')^2}{4 a^2 D^2 B} + \frac{B'}{a^2 r B^2} + \frac{C' B'}{2 a^2 C B^2} + \frac{D' B'}{4 a^2 D B^2} - \frac{C''}{a^2 C B} - \frac{D''}{2 a^2 D B} = \Lambda$$

$$R_{22}: \quad \frac{3 c^2 k}{a^2} + \frac{1}{a^2 r^2 C} - \frac{1}{a^2 r^2 B} + \Lambda - \frac{2 C'}{a^2 r C B} - \frac{D'}{2 a^2 r D B} - \frac{C' D'}{4 a^2 C D B} + \frac{B'}{2 a^2 r B^2} + \frac{C' B'}{4 a^2 C B^2} - \frac{C''}{2 a^2 C B} = \Lambda$$

$$R_{44}: \quad \frac{3 c^2 k}{a^2} + \Lambda - \frac{D'}{a^2 r D B} - \frac{C' D'}{2 a^2 C D B} + \frac{(D')^2}{4 a^2 D^2 B} + \frac{D' B'}{4 a^2 D B^2} - \frac{D''}{2 a^2 D B} = \Lambda$$

We can solve $R_{11}-2R_{22}-R_{44}$ algebraically
for B or Dsolve it for either C or D. We try the latter:

$R_{11}-2R_{22}-R_{44}$:

$$-\frac{6 c^2 k}{a^2} - \frac{2}{a^2 r^2 C} + \frac{2}{a^2 r^2 B} - 2 \Lambda + \frac{2 C'}{a^2 r C B} + \frac{(C')^2}{2 a^2 C^2 B} + \frac{2 D'}{a^2 r D B} + \frac{C' D'}{a^2 C D B} = -2 \Lambda$$

$$D(r) = e^{\int_1^r \frac{4 C^2 \left(-1+3 c^2 k B \rho^2\right) - \rho^2 C'[\rho]^2 + 4 C (B - \rho C'[\rho])}{2 C \rho (2 C + \rho C'[\rho])} d\rho} \alpha$$ where α is a constant of integration.

Because $g_{44}=-a^2 D(r) p^2$, α can be absorbed in the radius p:

$$D(r) = e^{\int_1^r \frac{4 C^2 \left(-1+3 c^2 k B \rho^2\right) - \rho^2 C'[\rho]^2 + 4 C (B - \rho C'[\rho])}{2 C \rho (2 C + \rho C'[\rho])} d\rho}$$

Plugging back in, the integral conveniently disappears from
all of the equations, R_{11} and R_{44} become too lengthy to show here, and

$$R_{22}: \quad \frac{3 c^2 k}{2 a^2} + \frac{1}{2 a^2 r^2 C} - \frac{1}{2 a^2 r^2 B} + \Lambda - \frac{3 C'}{2 a^2 r C B} + \frac{(C')^2}{8 a^2 C^2 B} + \frac{B'}{2 a^2 r B^2} + \frac{C' B'}{4 a^2 C B^2} - \frac{C''}{2 a^2 C B} = \Lambda$$

R_{11}-R_{44} is also somewhat lengthy, but if we put the LHS side over a common denominator, we can ignore the denominator because the RHS is zero. The resulting simplified equation is:

R_{11}-R_{44}: $r^2 B (C')^2 + 4 C^2 \left(-B + 3 c^2 k r^2 B^2 + r B'\right) + 2 C \left(2 B^2 + r^2 C' B' - 2 r B (3 C' + r C'')\right) = 0$

R_{11}-R_{44} can be solved for B:

$$B(r) = \frac{r (2 C + r C')^2}{\sqrt{C} \left(\beta - \int_1^r -\frac{2 \left(1+3 c^2 k C \rho^2\right) (2 C+\rho C'[\rho])}{\sqrt{C}} d\rho\right)}$$ where β is a constant of integration.

or $$B(\rho) = \frac{\rho (2 C + \rho C')^2}{\sqrt{C} \left(\beta - \int_1^\rho -\frac{2 \left(1+3 c^2 k C \sigma^2\right) (2 C+\sigma C'[\sigma])}{\sqrt{C}} d\sigma\right)}.$$

Mathematica note: In $D(r)$, above, Mathematica code uses K[1] for the integration dummy variable (which I change to ρ for better readability). In $B(r)$, just above, Mathematica again uses the default dummy variable K[1]. This causes a programming problem here because $B(\rho)$ appears in $D(r)$ expression. When $B(r)$ is plugged back into $D(r)$, the $D(r)$ expression will contain a double integral and Mathematica will use K[1] for both dummy variables of integration.

It is necesary to override the Mathematica default and make one of the dummy variables K[2]. For programming, it turns out to be easiest to just assign K[2] to the first function, $D(r)$, and then let the Mathematica assign the default K[1] to the second function, $B(r)$.

Further note: When an expression has just a single integral of r, I dispaly ρ as the dummy variable. When there is a double integral of r, I display σ as the inner dummy variable of integration and ρ as the outer dummy variable.

Plugging $B(\rho)$ into $D(r)$ yields

$$D(r) = e^{\int_1^r \frac{-\rho^2 C'[\rho]^2 + 4 C \left(-\rho C'[\rho] + \frac{\rho (2 C+\rho C'[\rho])^2}{\sqrt{C} \left(\beta - \int_1^\rho -\frac{2 \left(1+3 c^2 k C \sigma^2\right) (2 C+\sigma C'[\sigma])}{\sqrt{C}} d\sigma\right)}\right) + 4 C^2 \left(-1 + \frac{3 c^2 k \rho^3 (2 C+\rho C'[\rho])^2}{\sqrt{C} \left(\beta - \int_1^\rho -\frac{2 \left(1+3 c^2 k C \sigma^2\right) (2 C+\sigma C'[\sigma])}{\sqrt{C}} d\sigma\right)}\right)}{2 C \rho (2 C+\rho C'[\rho])} d\rho}$$

Amazingly, when we plug $B(r)$ and $D(r)$ back into the equations, the remaining equations become identities:

R_{11}: $\Lambda = \Lambda$

R_{22}: $\Lambda = \Lambda$

R_{44}: $\Lambda = \Lambda$

We have a solution to the Λ field equations. However, the formula for $D(r)$ can be simplified, completely eliminating the double integral! First, it can be rearranged:

$$D(r) = \text{Exp}\left[\int_1^r \left(-\frac{2 C[\rho]}{\rho (2 C[\rho] + \rho C'[\rho])} - \frac{2 C'[\rho]}{2 C[\rho] + \rho C'[\rho]} - \frac{\rho C'[\rho]^2}{2 C[\rho] (2 C[\rho] + \rho C'[\rho])}\right) d\rho\right.$$

$$\times\ \mathrm{Exp}\left[\int_1^r \left(\frac{2\left(1+3c^2 k\rho^2 C[\rho]\right)\left(2C[\rho]+\rho C'[\rho]\right)}{\sqrt{C[\rho]}\left(\beta+\int_1^\rho \frac{2\left(1+3c^2 k\sigma^2 C[\sigma]\right)\left(2C[\sigma]+\sigma C'[\sigma]\right)}{\sqrt{C[\sigma]}}\,d\sigma\right)}\right)d\rho\right]$$

Next, the first three terms can be combined and simplified:

$$\begin{aligned}
D(r) &= \mathrm{Exp}\left[\int_1^r \left(-\frac{1}{\rho}-\frac{C'[\rho]}{2C[\rho]}+\frac{2\left(1+3c^2 k\rho^2 C[\rho]\right)\left(2C[\rho]+\rho C'[\rho]\right)}{\sqrt{C[\rho]}\left(\beta+\int_1^\rho \frac{2\left(1+3c^2 k\sigma^2 C[\sigma]\right)\left(2C[\sigma]+\sigma C'[\sigma]\right)}{\sqrt{C[\sigma]}}\,d\sigma\right)}\right)d\rho\right] \\
&= \mathrm{Exp}\left[\int_1^r -\frac{1}{\rho}\,d\rho\right]\ \mathrm{Exp}\left[-\frac{1}{2}\int_1^r \frac{C'[\rho]}{C[\rho]}\,d\rho\right] \\
&\times\ \mathrm{Exp}\left[\int_1^r \left(\frac{2\left(1+3c^2 k\rho^2 C[\rho]\right)\left(2C[\rho]+\rho C'[\rho]\right)}{\sqrt{C[\rho]}\left(\beta+\int_1^\rho \frac{2\left(1+3c^2 k\sigma^2 C[\sigma]\right)\left(2C[\sigma]+\sigma C'[\sigma]\right)}{\sqrt{C[\sigma]}}\,d\sigma\right)}\right)d\rho\right]
\end{aligned}$$

Next:

$$\mathrm{Exp}\left[-\int_1^r \frac{1}{\rho}\,d\rho\right] = \frac{1}{r}$$

$$\mathrm{Exp}\left[-\frac{1}{2}\int_1^r \frac{C'[\rho]}{C[\rho]}\,d\rho\right] = \frac{1}{\sqrt{C[r]}}$$

$$D(r) = \frac{1}{r\sqrt{C[r]}}\,\mathrm{Exp}\left[\int_1^r \left(\frac{2\left(1+3c^2 k\rho^2 C[\rho]\right)\left(2C[\rho]+\rho C'[\rho]\right)}{\sqrt{C[\rho]}\left(\beta+\int_1^\rho \frac{2\left(1+3c^2 k\sigma^2 C[\sigma]\right)\left(2C[\sigma]+\sigma C'[\sigma]\right)}{\sqrt{C[\sigma]}}\,d\sigma\right)}\right)d\rho\right]$$

Now, let

$$f(r) \equiv \frac{2\left(1+3c^2 k r^2 C[r]\right)\left(2C[r]+\sigma C'[r]\right)}{\sqrt{C[r]}}$$

and

$$g(r) = \beta + \int f(r)\,dr.$$

$$\Rightarrow\ g'(r) = f(r)$$

$$\Rightarrow\ \int_1^r \frac{f[\rho]}{\beta+\int f[\rho]\,d\rho} = \int_1^r \frac{g'[\rho]}{g[\rho]}\,d\rho = \ln[g[r]]$$

$$\begin{aligned}
\Rightarrow\ D(r) &= \frac{1}{r\sqrt{C[r]}}\,\mathrm{Exp}\left[\int_1^r \left(\frac{2\left(1+3c^2 k\rho^2 C[\rho]\right)\left(2C[\rho]+\rho C'[\rho]\right)}{\sqrt{C[\rho]}\left(\beta+\int_1^\rho \frac{2\left(1+3c^2 k\sigma^2 C[\sigma]\right)\left(2C[\sigma]+\sigma C'[\sigma]\right)}{\sqrt{C[\sigma]}}\,d\sigma\right)}\right)d\rho\right] \\
&= \frac{1}{r\sqrt{C[r]}}\,\mathrm{Exp}\left[\int_1^r \frac{f[\rho]}{\beta+\int_1^\rho f[\sigma]\,d\sigma}\right] = \frac{1}{r\sqrt{C[r]}}\,\mathrm{Exp}\left[\int_1^r \frac{g'[\rho]}{g[\rho]}\,d\rho\right] \\
&= \frac{\mathrm{Exp}[\ln[g[r]]]}{r\sqrt{C[r]}} = \frac{g[r]}{r\sqrt{C[r]}} \\
&= \frac{\beta+\int f[r]\,dr}{r\sqrt{C[r]}} = \frac{1}{r\sqrt{C[r]}}\left(\beta+\int_1^r \frac{2\left(1+3c^2 k\rho^2 C[\rho]\right)\left(2C[\rho]+\rho C'[\rho]\right)}{\sqrt{C[\rho]}}\,d\rho\right)
\end{aligned}$$

$$\Rightarrow \quad \beta + \int_1^r \frac{2\left(1+3c^2 k \rho^2 C[\rho]\right)\left(2C[\rho]+\rho C'[\rho]\right)}{\sqrt{C[\rho]}} d\rho = r\sqrt{C[r]}\,D[r]$$

$$\Rightarrow \quad \frac{1}{B(r)} = \frac{\sqrt{C[r]}\left(\beta - \int_1^r -\frac{2\left(1+3c^2 k \rho^2 C[\rho]\right)\left(2C[\rho]+\rho C'[\rho]\right)}{\sqrt{C[\rho]}} d\rho\right)}{r\left(2C[r]+r C'[r]\right)^2}$$

$$= \frac{\sqrt{C[r]}\; r\sqrt{C[r]}\; D[r]}{r\left(2C[r]+r C'[r]\right)^2} = \frac{C[r]\,D[r]}{\left(2C[r]+r C'[r]\right)^2}$$

$$\Rightarrow \quad D[r] = \frac{\left(2C[r]+r C'[r]\right)^2}{B[r]\,C[r]}.$$

We have an exact solution to the Λ field equations:

$$A(t,r) = \mathcal{A}(t) = \frac{4c^2 a'[t]^2}{4c^2 k + \Lambda a[t]^2} \quad \text{where k is a free constant,}$$

$$B(t,r) = a(t)^2 B(r) = a[t]^2 \frac{r\left(2C[r]+r C'[r]\right)^2}{\sqrt{C[r]}\left(\beta - \int_1^r -\frac{2\left(1+3c^2 k \rho^2 C[\rho]\right)\left(2C[\rho]+\rho C'[\rho]\right)}{\sqrt{C[\rho]}} d\rho\right)}$$

where β is a free constant,

$$\mathbb{C}(t,r) = a[t]^2 C[r] \quad \text{where } C(r) \text{ is a free parameter,}$$

$$\mathbb{D}(t,r) = a(t)^2 D(r) = a(t)^2 \frac{\left(2C[r]+r C'[r]\right)^2}{B[r]\,C[r]}$$

and line element (using just the symbols for B and D

$$ds^2 = \frac{4c^2 \dot{a}[t]^2}{4c^2 k + \Lambda a[t]^2} dt^2 - a[t]^2 B[r]\, dr^2 - r^2 a[t]^2 C[r]\, d\theta^2 - r^2 a[t]^2 \mathrm{Sin}[\theta]^2 C[r]\, d\phi^2 - p^2 a[t]^2 D[r]\, d\psi^2$$

a-II Solve Case a for a(t) when A(t,r) = c^2

Solving $\frac{4c^2 \dot{a}[t]^2}{4c^2 k + \Lambda a[t]^2} = c^2$ yields

$$a(t) = \pm \frac{2c\sqrt{k}\,\mathrm{Sinh}\left[\frac{1}{2}\sqrt{\Lambda}\,(t+\chi)\right]}{\sqrt{\Lambda}} \quad \text{where } \chi \text{ is a free constant.}$$

Example of Case a: C(r)=1

For an example of this solution, let

$$C[r] = 1,$$

and replace β by $4(\gamma+1+c^2 k)$.

First,

$$\mathbb{C}(t,r) = a(t)^2\, C(r) = a[t]^2$$

Second, $\mathbb{B}(r)$ simplifies to

$$\mathbb{B}(r) = \frac{4\,r}{\beta - \int_1^r -4\left(1+3\,c^2\,k\,\rho^2\right)d\rho} = \frac{4\,r}{-4+4\,r+4\,c^2\,k\left(-1+r^3\right)+\beta}$$

$$= \frac{r}{r+c^2\,k\,r^3+\gamma}$$

and $\mathbb{B}(t,r) = a(t)^2\,\mathbb{B}(r) = \dfrac{r\,a[t]^2}{r+c^2\,k\,r^3+\gamma}$.

Third, then $\mathbb{D}(r)$ become

$$\mathbb{D}(r) = \frac{4\left(r+c^2\,k\,r^3+\gamma\right)}{r}$$

and $\mathbb{D}(t,r) = a(t)^2\mathbb{D}(r)$.

Thus, this solution to the Λ field equations is:

$$\mathbb{A}(t) = \frac{4\,c^2\,\dot{a}[t]^2}{4\,c^2\,k+\Lambda\,a[t]^2} \quad \text{where k is a free constant,}$$

$$\mathbb{B}(t,r) = a[t]^2\frac{r}{r+c^2\,k\,r^3+\gamma} \quad \text{where } \gamma \text{ is a free constant,}$$

$$\mathbb{C}(t,r) = a[t]^2,$$

and

$$\mathbb{D}(t,r) = a(t)^2\frac{4\left(r+c^2\,k\,r^3+\gamma\right)}{r}.$$

a_II: $\mathbb{B}(t,r)=a(t)^2\mathbb{B}(r)$, $\mathbb{C}(t,r)=a(t)^2\mathbb{C}(r)$, $\mathbb{A}(t,r)=c^2$

We solve Case a again, but with the extra condition that $\mathbb{A}(t,r)=c^2$.
While Case a has a solution containing $\mathbb{A}(t,r)=c^2$, it is possible we may obtain an alternate solution by setting $\mathbb{A}(t,r)=c^2$ before solving the ODEs.

The field equations become

$$R_{00}: \quad \frac{4\,\ddot{a}}{a} = \Lambda$$

$$R_{11}: \quad \frac{3\,\dot{a}^2}{a^2}+\frac{\ddot{a}}{a}-\frac{2\,\mathbb{C}'}{a^2\,r\,\mathbb{C}\,\mathbb{B}}+\frac{(\mathbb{C}')^2}{2\,a^2\,\mathbb{C}^2\,\mathbb{B}}+\frac{(\mathbb{D}')^2}{4\,a^2\,\mathbb{D}^2\,\mathbb{B}}+\frac{\mathbb{B}'}{a^2\,r\,\mathbb{B}^2}+\frac{\mathbb{C}'\,\mathbb{B}'}{2\,a^2\,\mathbb{C}\,\mathbb{B}^2}+\frac{\mathbb{D}'\,\mathbb{B}'}{4\,a^2\,\mathbb{D}\,\mathbb{B}^2}-\frac{\mathbb{C}''}{a^2\,\mathbb{C}\,\mathbb{B}}-\frac{\mathbb{D}''}{2\,a^2\,\mathbb{D}\,\mathbb{B}} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2\,\mathbb{C}}-\frac{1}{a^2\,r^2\,\mathbb{B}}+\frac{3\,\dot{a}^2}{a^2}+\frac{\ddot{a}}{a}-\frac{2\,\mathbb{C}'}{a^2\,r\,\mathbb{C}\,\mathbb{B}}-\frac{\mathbb{D}'}{2\,a^2\,r\,\mathbb{D}\,\mathbb{B}}-\frac{\mathbb{C}'\,\mathbb{D}'}{4\,a^2\,\mathbb{C}\,\mathbb{D}\,\mathbb{B}}+\frac{\mathbb{B}'}{2\,a^2\,r\,\mathbb{B}^2}+\frac{\mathbb{C}'\,\mathbb{B}'}{4\,a^2\,\mathbb{C}\,\mathbb{B}^2}-\frac{\mathbb{C}''}{2\,a^2\,\mathbb{C}\,\mathbb{B}} = \Lambda$$

$$R_{44}: \quad \frac{3\,\dot{a}^2}{a^2}+\frac{\ddot{a}}{a}-\frac{\mathbb{D}'}{a^2\,r\,\mathbb{D}\,\mathbb{B}}-\frac{\mathbb{C}'\,\mathbb{D}'}{2\,a^2\,\mathbb{C}\,\mathbb{D}\,\mathbb{B}}+\frac{(\mathbb{D}')^2}{4\,a^2\,\mathbb{D}^2\,\mathbb{B}}+\frac{\mathbb{D}'\,\mathbb{B}'}{4\,a^2\,\mathbb{D}\,\mathbb{B}^2}-\frac{\mathbb{D}''}{2\,a^2\,\mathbb{D}\,\mathbb{B}} = \Lambda$$

Solving R_{00} yields

$a(t) = e^{\frac{t\sqrt{\Lambda}}{2}}\alpha + e^{-\frac{t\sqrt{\Lambda}}{2}}\beta$ where α and β are constants of integration.

We can also algebraically solve $R_{11}-2R_{22}-R_{44}$ for $B(r)$:

$$R_{11}-2R_{22}-R_{44}: \quad -\frac{2}{a^2 r^2 C} + \frac{2}{a^2 r^2 B} - \frac{6\dot{a}^2}{a^2} - \frac{2\ddot{a}}{a} + \frac{2C'}{a^2 r C B} + \frac{(C')^2}{2a^2 C^2 B} + \frac{2D'}{a^2 r D B} + \frac{C'D'}{a^2 C D B} = -2\Lambda$$

$$B(r) = -\frac{(2C + rC')(2CD + rDC' + 2rCD')}{4CD\left(-1 + a^2 r^2 C\Lambda - 3r^2 C\dot{a}^2 - a r^2 C\ddot{a}\right)}$$

In order for $B(r)$ to strictly be a function of r, the t terms in the denominator must disappear. There are no terms in the numerator to cancel them out, so the denominator terms must be self-canceling. It turns out they are. Call the denominator expression in the parentheses $p(t,r)$:

$$p(t,r) = -1 + a^2 r^2 C\Lambda - 3r^2 C\dot{a}^2 - a r^2 C\ddot{a}$$

When we plug in the formula above for $a(t)$, this simplifies to

$$p(t,r) = -1 + 3r^2\alpha\beta\Lambda C[r]$$

$$\Rightarrow \quad B(r) = -\frac{(2C + rC')(2CD + rDC' + 2rCD')}{4CD\left(-1 + 3r^2 C\alpha\beta\Lambda\right)}$$, strictly a function of r.

When we plug $B(r)$ into the remaining equations, they are each a page long and I can't find an algebraic combination that simplifies them. However, the terms on the LHS of $R_{11}-R_{44}$ all have the same denominator, and the RHS is zero. So, we can ignore the denominators:

$$\begin{aligned}
&48 e^{t\sqrt{\Lambda}} r C^3 D^2 \alpha\beta\Lambda + 72 e^{t\sqrt{\Lambda}} r^2 C^2 D^2 \alpha\beta\Lambda C' + 36 e^{t\sqrt{\Lambda}} r^3 C D^2 \alpha\beta\Lambda (C')^2 + \\
&6 e^{t\sqrt{\Lambda}} r^4 D^2 \alpha\beta\Lambda (C')^3 + 16 e^{t\sqrt{\Lambda}} C^2 D D' + 8 e^{t\sqrt{\Lambda}} r C D C' D' + 24 e^{t\sqrt{\Lambda}} r^3 C^2 D \alpha\beta\Lambda C' D' + \\
&6 e^{t\sqrt{\Lambda}} r^2 D (C')^2 D' - 6 e^{t\sqrt{\Lambda}} r^4 C D \alpha\beta\Lambda (C')^2 D' - 4 e^{t\sqrt{\Lambda}} r^2 C D D' C'' + 12 e^{t\sqrt{\Lambda}} r^4 C^2 D \alpha\beta\Lambda D' C'' + \\
&8 e^{t\sqrt{\Lambda}} r C^2 D D'' - 24 e^{t\sqrt{\Lambda}} r^3 C^3 D \alpha\beta\Lambda D'' + 4 e^{t\sqrt{\Lambda}} r^2 C D C' D'' - 12 e^{t\sqrt{\Lambda}} r^4 C^2 D \alpha\beta\Lambda C' D'' \\
&= 0
\end{aligned}$$

This equation is still too difficult for me to solve, so I simplify it by setting the product $\alpha\beta=0$:

$$\begin{aligned}
&16 e^{t\sqrt{\Lambda}} C^2 D D' + 8 e^{t\sqrt{\Lambda}} r C D C' D' + 6 e^{t\sqrt{\Lambda}} r^2 D (C')^2 D' - \\
&4 e^{t\sqrt{\Lambda}} r^2 C D D' C'' + 8 e^{t\sqrt{\Lambda}} r C^2 D D'' + 4 e^{t\sqrt{\Lambda}} r^2 C D C' D'' \\
&= 0
\end{aligned}$$

$$\Rightarrow \quad C(r) = \frac{K}{r^2 (k + D)^2}$$

Plugging this back in, the remaining equations are satisfied:

R_{11}: $\Lambda = \Lambda$

R_{22}: $\Lambda = \Lambda$

R_{44}: $\Lambda = \Lambda$

As hoped, we have found a different exact solution to the Λ field equations than that found in Case a:

$a(t) = e^{\frac{t\sqrt{\Lambda}}{2}}\alpha$ where α is a free constant,

$A(t,r) = c^2$,

$B(t,r) = a(t)^2 B(r) = a(t)^2 - \frac{k K D'[r]^2}{D[r]\,(k + D[r])^4}$ where k and K are free constants,

$C(t,r) = \frac{e^{t\sqrt{\Lambda}}\alpha^2 K}{r^2\,(k + D[r])^2}$,

$D(t,r) = a(t)^2 D(r)$ where $D(r)$ is a free parameter,

and line element

$$ds^2 = c^2\,dt^2 + \frac{e^{t\sqrt{\Lambda}} k\,\alpha^2 K D'[r]^2}{D[r]\,(k + D[r])^4}\,dr^2 - \frac{e^{t\sqrt{\Lambda}}\alpha^2 K}{(k + D[r])^2}\,d\theta^2 - \frac{e^{t\sqrt{\Lambda}}\alpha^2 K\,\mathrm{Sin}[\theta]^2}{(k + D[r])^2}\,d\phi^2 - e^{t\sqrt{\Lambda}} p^2 \alpha^2 D[r]\,d\psi^2$$

We obtained this solution by arbitrarily setting the product $\alpha\beta=0$. We can possibly obtain other exact solutions. Another natural choice was to use is $\alpha=-\beta=\frac{1}{2}$, but that did not lead to a solvable equation.

b: B(t,r)=B(r), C(t,r)=C(r)

I am able below to solve (with difficulty) for A and B and reduce the field equations down to 2 equations in 2 unknowns, but I couldn't proceed any further. The reader may wish to skip to the next section.

The field equations begin with:

R_{00}: $$\frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{A'}{A\,r\,B} + \frac{(A')^2}{4\,A^2\,B} - \frac{A'\,C'}{2\,A\,C\,B} - \frac{A'\,D'}{4\,A\,D\,B} + \frac{A'\,B'}{4\,A\,B^2} - \frac{A''}{2\,A\,B} = \Lambda$$

R_{01}: $$-\frac{\dot{a}\,A'}{2\,a\,A} + \frac{\dot{a}\,D'}{2\,a\,D} = 0$$

R_{11}: $$\frac{(A')^2}{4\,A^2\,B} - \frac{2\,C'}{r\,C\,B} + \frac{(C')^2}{2\,C^2\,B} + \frac{(D')^2}{4\,D^2\,B} + \frac{B'}{r\,B^2} + \frac{A'\,B'}{4\,A\,B^2} + \frac{C'\,B'}{2\,C\,B^2} + \frac{D'\,B'}{4\,D\,B^2} - \frac{A''}{2\,A\,B} - \frac{C''}{C\,B} - \frac{D''}{2\,D\,B} = \Lambda$$

R_{22}: $$\frac{1}{r^2\,C} - \frac{1}{r^2\,B} - \frac{A'}{2\,A\,r\,B} - \frac{2\,C'}{r\,C\,B} - \frac{A'\,C'}{4\,A\,C\,B} - \frac{D'}{2\,r\,D\,B} - \frac{C'\,D'}{4\,C\,D\,B} + \frac{B'}{2\,r\,B^2} + \frac{C'\,B'}{4\,C\,B^2} - \frac{C''}{2\,C\,B} = \Lambda$$

R_{44}: $$\frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{D'}{r\,D\,B} - \frac{A'\,D'}{4\,A\,D\,B} - \frac{C'\,D'}{2\,C\,D\,B} + \frac{(D')^2}{4\,D^2\,B} + \frac{D'\,B'}{4\,D\,B^2} - \frac{D''}{2\,D\,B} = \Lambda$$

R_{01} can be simplified and then solved for A:

$$R_{01}: \quad \frac{A'}{A} - \frac{\mathcal{D}'}{\mathcal{D}} = 0$$

$A(t,r) = \mathcal{A}[t] \times \mathcal{D}[r]$ where $\mathcal{A}(t)$ is a new unknown.

The equations now become

$$R_{00}: \quad \frac{c^2 \ddot{a}}{a \mathcal{A} \mathcal{D}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2 \mathcal{D}} - \frac{\mathcal{D}'}{r \mathcal{D} B} - \frac{C' \mathcal{D}'}{2 C \mathcal{D} B} + \frac{\mathcal{D}' B'}{4 \mathcal{D} B^2} - \frac{\mathcal{D}''}{2 \mathcal{D} B} = \Lambda$$

$$R_{11}: \quad -\frac{2 C'}{r C B} + \frac{(C')^2}{2 C^2 B} + \frac{(\mathcal{D}')^2}{2 \mathcal{D}^2 B} + \frac{B'}{r B^2} + \frac{C' B'}{2 C B^2} + \frac{\mathcal{D}' B'}{2 \mathcal{D} B^2} - \frac{C''}{C B} - \frac{\mathcal{D}''}{\mathcal{D} B} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2 C} - \frac{1}{r^2 B} - \frac{2 C'}{r C B} - \frac{\mathcal{D}'}{r \mathcal{D} B} - \frac{C' \mathcal{D}'}{2 C \mathcal{D} B} + \frac{B'}{2 r B^2} + \frac{C' B'}{4 C B^2} - \frac{C''}{2 C B} = \Lambda$$

$$R_{44}: \quad \frac{c^2 \ddot{a}}{a \mathcal{A} \mathcal{D}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2 \mathcal{D}} - \frac{\mathcal{D}'}{r \mathcal{D} B} - \frac{C' \mathcal{D}'}{2 C \mathcal{D} B} + \frac{\mathcal{D}' B'}{4 \mathcal{D} B^2} - \frac{\mathcal{D}''}{2 \mathcal{D} B} = \Lambda$$

R_{44} has become redundant, the same as R_{00}.

$2R_{00}-R_{11}+2R_{22}$ can be algebraically solved for B:

$$2R_{00}-R_{11}+2R_{22}: \quad \frac{2}{r^2 C} - \frac{2}{r^2 B} + \frac{2 c^2 \ddot{a}}{a \mathcal{A} \mathcal{D}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{a \mathcal{A}^2 \mathcal{D}} - \frac{2 C'}{r C B} - \frac{(C')^2}{2 C^2 B} - \frac{4 \mathcal{D}'}{r \mathcal{D} B} - \frac{2 C' \mathcal{D}'}{C \mathcal{D} B} - \frac{(\mathcal{D}')^2}{2 \mathcal{D}^2 B} = 3 \Lambda$$

Unfortunately, the formula for $B(r)$ is lengthy and makes the equations much worse when plugging back in, so we try a different approach.

Instead of solving $2R_{00}-R_{11}+2R_{22}$, we can solve R_{00} for $\mathcal{A}$ or B (but not $\mathcal{D}$). Solving for B results in unsolvable equations. However, solving for $\mathcal{A}$ is more productive:

$$\mathcal{A}(t) = \frac{4 c^2 r C B^2 \dot{a}^2}{4 c^2 r C B^2 \gamma + 4 a^2 r C \mathcal{D} B^2 \Lambda + 4 a^2 C B \mathcal{D}' + 2 a^2 r B C' \mathcal{D}' - a^2 r C \mathcal{D}' B' + 2 a^2 r C B \mathcal{D}''}$$

where γ is a constant of integration

Observe that the numerator divided by $r\ C\ B^2$ is strictly a function of t. In order for the whole expression to be strictly a function of t, the denominator divided by $r\ C\ B^2$ must also equal some function of t, say $f(t)$.

$$\frac{\text{Denominator}}{r C B^2} = f(t): \quad 4 c^2 \gamma + 4 a^2 \mathcal{D} \Lambda + \frac{4 a^2 \mathcal{D}'}{r B} + \frac{2 a^2 C' \mathcal{D}'}{C B} - \frac{a^2 \mathcal{D}' B'}{B^2} + \frac{2 a^2 \mathcal{D}''}{B} = f[t]$$

We can solve this equation for $B(r)$:

$$B[r] = \frac{r^4 C^2 (D')^2}{\alpha - \int_1^r \frac{C^2 (-f + 4 c^2 \gamma + 4 a^2 D \Lambda) \rho^4 D'}{a^2} d\rho}$$ where α is a constant of integration.

This formula for $B(r)$ makes the equations unwieldy when it is plugged back in, so we don't use it. It's existence is sufficient to show that the equation

$$\frac{\text{Denominator}}{r C B^2} = f(t)$$

is solvable. This means that $\mathcal{A}(t)$ can (again) be compactly expressed as

$$\mathcal{A}(t) = \frac{\text{Numerator}}{\text{Denominator}} = \frac{4 c^2 \dot{a}^2}{f}.$$

Plugging $\mathcal{A}$ back into the equations should make R_{00} into the identity, 0=0, but it doesn't because we have suppessed the formula for B:

$$R_{00}: \quad \frac{\dot{f}}{8 a D \dot{a}} - \frac{D'}{r D B} - \frac{C' D'}{2 C D B} + \frac{D' B'}{4 D B^2} - \frac{D''}{2 D B} = \Lambda.$$

We can restore R_{00} as an identity if we eliminate t from R_{00} and then re-solve for $B(r)$. We eliminate t by solving $\dot{f} = \kappa\, a\, \dot{a}$, where κ is a proportionality constant:

$$f(t) = \zeta + \frac{1}{2} \kappa\, a[t]^2$$ where ζ is a constant of integration.

$\mathcal{A}(t)$ is now clearly seen to be a strict function of t,

$$\mathcal{A}(t) = \frac{4 c^2 \dot{a}[t]^2}{\zeta + \frac{1}{2} \kappa\, a[t]^2},$$

and R_{00} becomes

$$R_{00}: \quad \frac{\kappa}{8 D} - \frac{D'}{r D B} - \frac{C' D'}{2 C D B} + \frac{D' B'}{4 D B^2} - \frac{D''}{2 D B} = \Lambda.$$

When we re-solve R_{00} for $B(r)$, we obtain the desired simplified $B(r)$ formula, one that has constant κ instead of the $f(t)$ function:

$$B[r] = \frac{r^4 C^2 (D')^2}{\alpha - \int_1^r -\frac{1}{2} C^2 (\kappa - 8 D \Lambda) \rho^4 D' d\rho}$$ where α and κ are constants of integration.

Plugging this simplified version of B into the equations shows that R_{00} is indeed an identity and, in fact, so is R_{44}.

R_{00}: $\Lambda = \Lambda$

R_{44}: $\Lambda = \Lambda$

However, R_{11} and R_{22} have many terms and also include the $\mathbb{B}(r)$ integral. The simplest combination of the equations is $R_{11}-2R_{22}$ and even it has 6 terms involving the integral plus 8 more terms.

I reduced this to 2 equations in 2 unknowns but was unable to solve the case. In order to solve $R_{11}-2R_{22}$, I require one more simplification, $\mathbb{C}(r)=1$, which is Case bi, analyzed below after Case b_II, and it is solved starting from scratch.

b_Iα: $\mathbb{B}(t,r)=\mathbb{B}(r)$, $\mathbb{C}(t,r)=\mathbb{C}(r)$, $\mathbb{A}(t,r)=a(t)^2$

Restarting Case b from scratch using these constraints, the field equations become:

$$R_{00}:\quad -\frac{c^2 (a')^2}{a^4}+\frac{c^2 a''}{a^3} = \Lambda$$

$$R_{11}:\quad -\frac{2\,\mathbb{C}'}{r\,\mathbb{C}\,\mathbb{B}}+\frac{(\mathbb{C}')^2}{2\,\mathbb{C}^2\,\mathbb{B}}+\frac{(\mathbb{D}')^2}{4\,\mathbb{D}^2\,\mathbb{B}}+\frac{\mathbb{B}'}{r\,\mathbb{B}^2}+\frac{\mathbb{C}'\,\mathbb{B}'}{2\,\mathbb{C}\,\mathbb{B}^2}+\frac{\mathbb{D}'\,\mathbb{B}'}{4\,\mathbb{D}\,\mathbb{B}^2}-\frac{\mathbb{C}''}{\mathbb{C}\,\mathbb{B}}-\frac{\mathbb{D}''}{2\,\mathbb{D}\,\mathbb{B}} = \Lambda$$

$$R_{22}:\quad \frac{1}{r^2\,\mathbb{C}}-\frac{1}{r^2\,\mathbb{B}}-\frac{2\,\mathbb{C}'}{r\,\mathbb{C}\,\mathbb{B}}-\frac{\mathbb{D}'}{2\,r\,\mathbb{D}\,\mathbb{B}}-\frac{\mathbb{C}'\,\mathbb{D}'}{4\,\mathbb{C}\,\mathbb{D}\,\mathbb{B}}+\frac{\mathbb{B}'}{2\,r\,\mathbb{B}^2}+\frac{\mathbb{C}'\,\mathbb{B}'}{4\,\mathbb{C}\,\mathbb{B}^2}-\frac{\mathbb{C}''}{2\,\mathbb{C}\,\mathbb{B}} = \Lambda$$

$$R_{44}:\quad -\frac{c^2 (a')^2}{a^4}-\frac{\mathbb{D}'}{r\,\mathbb{D}\,\mathbb{B}}-\frac{\mathbb{C}'\,\mathbb{D}'}{2\,\mathbb{C}\,\mathbb{D}\,\mathbb{B}}+\frac{(\mathbb{D}')^2}{4\,\mathbb{D}^2\,\mathbb{B}}+\frac{\mathbb{D}'\,\mathbb{B}'}{4\,\mathbb{D}\,\mathbb{B}^2}+\frac{c^2 a''}{a^3}-\frac{\mathbb{D}''}{2\,\mathbb{D}\,\mathbb{B}} = \Lambda$$

R_{01} does not appear because we have assumed $\mathbb{B}(t,r) = \mathbb{B}(r)$, which is the solution to R_{01}.

We can solve R_{00} for $a(t)$:

$$a(t) = \pm\frac{i\,c\,K\,\mathrm{Sech}[\zeta + t\,K]}{\sqrt{\Lambda}}$$ where K and ζ are constants of integration.

This changes the first term of R_{44} so that we can easily solve it for $\mathbb{B}$:

$$\mathrm{lhs}[4,4] = \Lambda-\frac{\mathbb{D}'}{r\,\mathbb{D}\,\mathbb{B}}-\frac{\mathbb{C}'\,\mathbb{D}'}{2\,\mathbb{C}\,\mathbb{D}\,\mathbb{B}}+\frac{(\mathbb{D}')^2}{4\,\mathbb{D}^2\,\mathbb{B}}+\frac{\mathbb{D}'\,\mathbb{B}'}{4\,\mathbb{D}\,\mathbb{B}^2}-\frac{\mathbb{D}''}{2\,\mathbb{D}\,\mathbb{B}}$$

$$\mathbb{B}(r) = \frac{r^4\,\mathbb{C}^2\,\alpha\,(\mathbb{D}')^2}{\mathbb{D}}$$ where α is a constant of integration.

R_{00} and R_{44} become identities and the remaining equations become

$$R_{11}:\quad \frac{4\,\mathbb{D}}{r^6\,\mathbb{C}^2\,\alpha\,(\mathbb{D}')^2}+\frac{2\,\mathbb{D}\,\mathbb{C}'}{r^5\,\mathbb{C}^3\,\alpha\,(\mathbb{D}')^2}+\frac{3\,\mathbb{D}\,(\mathbb{C}')^2}{2\,r^4\,\mathbb{C}^4\,\alpha\,(\mathbb{D}')^2}-\frac{\mathbb{D}\,\mathbb{C}''}{r^4\,\mathbb{C}^3\,\alpha\,(\mathbb{D}')^2}+\frac{2\,\mathbb{D}\,\mathbb{D}''}{r^5\,\mathbb{C}^2\,\alpha\,(\mathbb{D}')^3}+\frac{\mathbb{D}\,\mathbb{C}'\,\mathbb{D}''}{r^4\,\mathbb{C}^3\,\alpha\,(\mathbb{D}')^3} = \Lambda$$

$$R_{22}:\quad \frac{1}{r^2\,\mathbb{C}}+\frac{\mathbb{D}}{r^6\,\mathbb{C}^2\,\alpha\,(\mathbb{D}')^2}+\frac{\mathbb{D}\,(\mathbb{C}')^2}{2\,r^4\,\mathbb{C}^4\,\alpha\,(\mathbb{D}')^2}-$$
$$\frac{1}{r^5\,\mathbb{C}^2\,\alpha\,\mathbb{D}'}-\frac{\mathbb{C}'}{2\,r^4\,\mathbb{C}^3\,\alpha\,\mathbb{D}'}-\frac{\mathbb{D}\,\mathbb{C}''}{2\,r^4\,\mathbb{C}^3\,\alpha\,(\mathbb{D}')^2}+\frac{\mathbb{D}\,\mathbb{D}''}{r^5\,\mathbb{C}^2\,\alpha\,(\mathbb{D}')^3}+\frac{\mathbb{D}\,\mathbb{C}'\,\mathbb{D}''}{2\,r^4\,\mathbb{C}^3\,\alpha\,(\mathbb{D}')^3} = \Lambda$$

However, I am unable to solve either of these equations or algebraic combinations of them.

b_II: B(t,r)=B(r), C(t,r)=C(r), A(t,r)=c²

We try Case b with the additional assumption that A(t,r)=c². Starting fresh, the field equations are:

$$R_{00}: \quad \frac{\ddot{a}}{a} = \Lambda$$

$$R_{01}: \quad \frac{\dot{a}\,D'}{2\,a\,D} = 0$$

$$R_{11}: \quad -\frac{2\,C'}{r\,C\,B} + \frac{(C')^2}{2\,C^2\,B} + \frac{(D')^2}{4\,D^2\,B} + \frac{B'}{r\,B^2} + \frac{C'\,B'}{2\,C\,B^2} + \frac{D'\,B'}{4\,D\,B^2} - \frac{C''}{C\,B} - \frac{D''}{2\,D\,B} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2\,C} - \frac{1}{r^2\,B} - \frac{2\,C'}{r\,C\,B} - \frac{D'}{2\,r\,D\,B} - \frac{C'\,D'}{4\,C\,D\,B} + \frac{B'}{2\,r\,B^2} + \frac{C'\,B'}{4\,C\,B^2} - \frac{C''}{2\,C\,B} = \Lambda$$

$$R_{44}: \quad \frac{\ddot{a}}{a} - \frac{D'}{r\,D\,B} - \frac{C'\,D'}{2\,C\,D\,B} + \frac{(D')^2}{4\,D^2\,B} + \frac{D'\,B'}{4\,D\,B^2} - \frac{D''}{2\,D\,B} = \Lambda$$

The solution to R_{00} is

$$a(t) = e^{t\sqrt{\Lambda}}\,\alpha + e^{-t\sqrt{\Lambda}}\,\beta$$ where α and β are constants of integration.

Since the denominator of R_{01} can be ignored and since $a \neq$ constant, then $\dot{a} \neq 0$, and we get that

$$D(r) = K,$$ a constant.

Since $g_{44} = -a(t)^2 D(r) p^2$, we can roll up K into the radius p:

$$D(r) = 1.$$

Plugging back in, R_{00} and R_{01} are satisfied and can be ignored. We see that R_{44} has also become satisfied:

$$R_{11}: \quad -\frac{2\,C'}{r\,C\,B} + \frac{(C')^2}{2\,C^2\,B} + \frac{B'}{r\,B^2} + \frac{C'\,B'}{2\,C\,B^2} - \frac{C''}{C\,B} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2\,C} - \frac{1}{r^2\,B} - \frac{2\,C'}{r\,C\,B} + \frac{B'}{2\,r\,B^2} + \frac{C'\,B'}{4\,C\,B^2} - \frac{C''}{2\,C\,B} = \Lambda$$

$$R_{44}: \quad \Lambda = \Lambda$$

We can algebraically solve $R_{11}-2R_{22}$ for B:

$$R_{11}-2R_{22}: \quad -\frac{2}{r^2\,C} + \frac{2}{r^2\,B} + \frac{2\,C'}{r\,C\,B} + \frac{(C')^2}{2\,C^2\,B} = -\Lambda$$

$$B(r) = -\frac{(2\,C[r] + r\,C'[r])^2}{2\,C[r]\,\left(-2 + r^2\,\Lambda\,C[r]\right)}$$

We see that the remaining equations are both solved:

R_{11}: $\Lambda = \Lambda$

R_{22}: $\Lambda = \Lambda$

We have found an exact solution to the Λ field equations:

$a(t) = e^{t\sqrt{\Lambda}}\alpha + e^{-t\sqrt{\Lambda}}\beta$ where α and β are free parameters,

$A(t,r) = c^2$,

$$B(t,r) = -\frac{(2\,\mathbb{C}[r] + r\,\mathbb{C}'[r])^2}{2\,\mathbb{C}[r]\left(-2 + r^2\Lambda\,\mathbb{C}[r]\right)},$$

$\mathbb{C}(t,r) = \mathbb{C}[r]$, a free parameter,

$\mathbb{D}(t,r) = a(t)^2$,

and line element

$$ds^2 = c^2\,dt^2 + \frac{(2\,\mathbb{C}[r] + r\,\mathbb{C}'[r])^2}{2\,\mathbb{C}[r]\left(-2 + r^2\Lambda\,\mathbb{C}[r]\right)}\,dr^2 - r^2\,\mathbb{C}[r]\,d\theta^2 - r^2\,\mathrm{Sin}[\theta]^2\,\mathbb{C}[r]\,d\phi^2 - p^2\left(e^{t\sqrt{\Lambda}}\alpha + e^{-t\sqrt{\Lambda}}\beta\right)^2 d\psi^2$$

bii: B(t,r)=B(r), $\mathbb{C}$(t,r)=$\mathbb{C}$(r)=1

The solution below involves setting $\mathbb{D}(r) = \frac{\beta}{2}r^2$, an educated guess.

This is not rigorous, so there is no guarantee that there are not other, more general, solutions to this case. Nonetheless, this solution is distinct from the prior case, which is also a subcase of Case b.

Following the Case b methodology, but using this simpler value of $\mathbb{C}(r)$, starting from scratch, the field equations become:

R_{00}: $$\frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{A'}{A\,r\,B} + \frac{(A')^2}{4\,A^2\,B} - \frac{A'\,\mathbb{D}'}{4\,A\,\mathbb{D}\,B} + \frac{A'\,B'}{4\,A\,B^2} - \frac{A''}{2\,A\,B} = \Lambda$$

R_{01}: $$-\frac{\dot{a}\,A'}{2\,a\,A} + \frac{\dot{a}\,\mathbb{D}'}{2\,a\,\mathbb{D}} = 0$$

R_{11}: $$\frac{(A')^2}{4\,A^2\,B} + \frac{(\mathbb{D}')^2}{4\,\mathbb{D}^2\,B} + \frac{B'}{r\,B^2} + \frac{A'\,B'}{4\,A\,B^2} + \frac{\mathbb{D}'\,B'}{4\,\mathbb{D}\,B^2} - \frac{A''}{2\,A\,B} - \frac{\mathbb{D}''}{2\,\mathbb{D}\,B} = \Lambda$$

R_{22}: $$\frac{1}{r^2} - \frac{1}{r^2\,B} - \frac{A'}{2\,A\,r\,B} - \frac{\mathbb{D}'}{2\,r\,\mathbb{D}\,B} + \frac{B'}{2\,r\,B^2} = \Lambda$$

R_{44}: $$\frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{\mathbb{D}'}{r\,\mathbb{D}\,B} - \frac{A'\,\mathbb{D}'}{4\,A\,\mathbb{D}\,B} + \frac{(\mathbb{D}')^2}{4\,\mathbb{D}^2\,B} + \frac{\mathbb{D}'\,B'}{4\,\mathbb{D}\,B^2} - \frac{\mathbb{D}''}{2\,\mathbb{D}\,B} = \Lambda$$

As in Case b, we solve R_{01}:

$A(t,r) = \mathcal{A}[t] \times \mathbb{D}[r]$ where $\mathcal{A}(t)$ is a new unknown.

Plugging back in:

$$R_{00}: \quad \frac{c^2 \ddot{a}}{a \mathcal{A} \mathcal{D}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2 \mathcal{D}} - \frac{\mathcal{D}'}{r \mathcal{D} \mathcal{B}} + \frac{\mathcal{D}' \mathcal{B}'}{4 \mathcal{D} \mathcal{B}^2} - \frac{\mathcal{D}''}{2 \mathcal{D} \mathcal{B}} = \Lambda$$

$$R_{11}: \quad \frac{(\mathcal{D}')^2}{2 \mathcal{D}^2 \mathcal{B}} + \frac{\mathcal{B}'}{r \mathcal{B}^2} + \frac{\mathcal{D}' \mathcal{B}'}{2 \mathcal{D} \mathcal{B}^2} - \frac{\mathcal{D}''}{\mathcal{D} \mathcal{B}} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2} - \frac{1}{r^2 \mathcal{B}} - \frac{\mathcal{D}'}{r \mathcal{D} \mathcal{B}} + \frac{\mathcal{B}'}{2 r \mathcal{B}^2} = \Lambda$$

$$R_{44}: \quad \frac{c^2 \ddot{a}}{a \mathcal{A} \mathcal{D}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2 \mathcal{D}} - \frac{\mathcal{D}'}{r \mathcal{D} \mathcal{B}} + \frac{\mathcal{D}' \mathcal{B}'}{4 \mathcal{D} \mathcal{B}^2} - \frac{\mathcal{D}''}{2 \mathcal{D} \mathcal{B}} = \Lambda$$

R_{44} has become redundant, the same as R_{00}.

We can solve R_{00} for $\mathcal{A}$:

$$\mathcal{A}(t) = \frac{4 c^2 r \mathcal{B}^2 \dot{a}^2}{4 a^2 r \mathcal{D} \mathcal{B}^2 \Lambda + 4 c^2 r \mathcal{B}^2 \mathbb{c}_1 + 4 a^2 \mathcal{B} \mathcal{D}' - a^2 r \mathcal{D}' \mathcal{B}' + 2 a^2 r \mathcal{B} \mathcal{D}''}$$

where $\mathbb{c}_1$ is a constant of integration.

The numerator divided by $r\ \mathcal{B}^2$ is strictly a function of t, ever so slightly different than Case b. In order for the whole expression to be strictly a function of t, the denominator divided by $r\ \mathcal{B}^2$ must equal some function of t, say $f(t)$.

$$\frac{\text{Denominator}}{r \mathcal{B}^2} = f(t): \quad 4 a^2 \mathcal{D} \Lambda + 4 c^2 \mathbb{c}_1 + \frac{4 a^2 \mathcal{D}'}{r \mathcal{B}} - \frac{a^2 \mathcal{D}' \mathcal{B}'}{\mathcal{B}^2} + \frac{2 a^2 \mathcal{D}''}{\mathcal{B}} = f[t]$$

We can solve this equation for $\mathcal{B}(r)$, getting the same result as in Case b:

$$\mathcal{B}[r] = \frac{r^4 (\mathcal{D}')^2}{\alpha - \int_1^r \frac{\rho^4 \left(-f + 4 a^2 \mathcal{D} \Lambda + 4 c^2 \mathbb{c}_1\right) \mathcal{D}'}{a^2} \, d\rho}$$

where α and $\mathbb{c}_1$ are constants of integration.

This formula for $\mathcal{B}(r)$ shows that the equation

$$\frac{\text{Denominator}}{r \mathcal{B}^2} = f(t)$$

is solvable, meaning that $\mathcal{A}(t)$ can (again) be compactly expressed as

$$\mathcal{A}(t) = \frac{\text{Numerator}}{\text{Denominator}} = \frac{4 c^2 \dot{a}^2}{f}.$$

Plugging $\mathcal{A}$ and $\mathcal{B}$ back into the equations is unwieldy because of $\mathcal{B}$, but we are not interested in the formula for $\mathcal{B}$ just now. Rather, we are focused on just using the above simple expression for $\mathcal{A}(t)$.

Plugging just $\mathcal{A}(t)$ back into the equations should make R_{00} into the identity, 0=0, but it doesn't because we have suppessed the formula for $\mathbb{B}$:

$$R_{00}: \quad \frac{\dot{f}}{8\,a\,\mathbb{D}\,\dot{a}} - \frac{\mathbb{D}'}{r\,\mathbb{D}\,\mathbb{B}} + \frac{\mathbb{D}'\,\mathbb{B}'}{4\,\mathbb{D}\,\mathbb{B}^2} - \frac{\mathbb{D}''}{2\,\mathbb{D}\,\mathbb{B}}.$$

As in Case b, we can make R_{00} an identity by eliminating the t terms, i.e., solving

$\dot{f} = \kappa\, a\, \dot{a}$ where κ is a proportionality constant,

and then re-solving R_{00} for $\mathbb{B}$.

Solving the ODE, above, for f yields:

$f(t) = \zeta + \frac{1}{2}\kappa\, a[t]^2$ where ζ is a constant of integration.

$\mathcal{A}(t)$ then becomes

$$\mathcal{A}(t) = \frac{4\,c^2\,\dot{a}[t]^2}{\zeta + \frac{1}{2}\kappa\, a[t]^2},$$

and R_{00} becomes

$$R_{00}: \quad \frac{\kappa}{8\,\mathbb{D}} - \frac{\mathbb{D}'}{r\,\mathbb{D}\,\mathbb{B}} + \frac{\mathbb{D}'\,\mathbb{B}'}{4\,\mathbb{D}\,\mathbb{B}^2} - \frac{\mathbb{D}''}{2\,\mathbb{D}\,\mathbb{B}}.$$

When we re-solve R_{00} for $\mathbb{B}(r)$, we obtain a simplified version of the $\mathbb{B}(r)$ formula, above, that has constant κ instead of the $f(t)$ function:

$$\mathbb{B}[r] = \frac{r^4\,(\mathbb{D}')^2}{\alpha - \int_1^r -\frac{1}{2}\,(\kappa - 8\,\mathbb{D}\,\Lambda)\,\rho^4\,\mathbb{D}'\,d\rho}$$ where α and κ are constants of integration.

Plugging this simplified version of $\mathbb{B}$ into the equations shows that R_{00} is indeed an identity and, in fact, so is R_{44}.

R_{00}: $\Lambda = \Lambda$

R_{44}: $\Lambda = \Lambda$

R_{11} and R_{22}, however, are rather messy. The simpler of the two is R_{22}:

$$R_{22}: \quad \frac{1}{r^2} + \frac{\alpha}{r^6\,(\mathbb{D}')^2} - \frac{\alpha}{r^5\,\mathbb{D}\,\mathbb{D}'} - \frac{\kappa}{4\,r\,\mathbb{D}'} + \frac{2\,\mathbb{D}\,\Lambda}{r\,\mathbb{D}'} + \frac{\alpha\,\mathbb{D}''}{r^5\,(\mathbb{D}')^3} - \frac{\int_1^r -\frac{1}{2}\,(\kappa - 8\,\mathbb{D}\,\Lambda)\,\rho^4\,\mathbb{D}'\,d\rho}{r^6\,(\mathbb{D}')^2} + \frac{\int_1^r -\frac{1}{2}\,(\kappa - 8\,\mathbb{D}\,\Lambda)\,\rho^4\,\mathbb{D}'\,d\rho}{r^5\,\mathbb{D}\,\mathbb{D}'} - \frac{\mathbb{D}''\int_1^r -\frac{1}{2}\,(\kappa - 8\,\mathbb{D}\,\Lambda)\,\rho^4\,\mathbb{D}'\,d\rho}{r^5\,(\mathbb{D}')^3} = \Lambda$$

I am unable solve these equations or any combination of them using standard methods. The simplest combination is R_{11}-$2R_{22}$ but it is still quite complex.

However, using a hunch, I am able to magically find a solution by making the denominators of the three terms that contain the integral proportional to each other. That is, the first two denominators are $r^5 \mathcal{D}'[r]^3$ and $r^6 \mathcal{D}'[r]^2$, so we want $r^5 \mathcal{D}'[r]^3 = \beta\, r^6\, \mathcal{D}'[r]^2$ for some proportionality constant β, or $\mathcal{D}'(r) = \beta\, r \;\Rightarrow\; \mathcal{D}(r) = \frac{\beta}{2} r^2 + K$ where K is a constant of integration.

Similarly, the 3rd denominator is $r^5 \mathcal{D}(r) \mathcal{D}'(r)$ and we want $\beta r^6 \mathcal{D}'(r)^2 = \gamma r^5 \mathcal{D}(r) \mathcal{D}'(r)$ for some proportionality constant γ. Combining these, it is straightforward to show that $K = 0$, so that

$$\mathcal{D}(r) = \frac{\beta}{2} r^2$$

$$\Rightarrow \quad \mathcal{D}'(r) = r\,\beta$$

$$\Rightarrow \quad \mathcal{B}(r) = \frac{r^6 \beta^2}{\alpha - \int_1^r -\frac{1}{2} \beta \rho^5 \left(\kappa - 4 \beta \Lambda \rho^2\right) d\rho} = -\frac{12\, r^6 \beta^2}{-12\,\alpha + \beta \left(\kappa - r^6 \kappa + 3 \left(-1 + r^8\right) \beta \Lambda\right)}.$$

When we plug this formula into R_{11} and R_{22} we get

$$R_{11}: \quad \frac{12\,\alpha}{r^8 \beta^2} - \frac{\kappa}{r^8 \beta} + \Lambda + \frac{3\,\Lambda}{r^8} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2} - \frac{\kappa}{4\, r^2 \beta} + \Lambda = \Lambda$$

R_{22} is satisfied when

$$\beta = \frac{\kappa}{4}$$

$$\Rightarrow \quad \mathcal{D}(r) = \frac{r^2 \kappa}{8}$$

$$\Rightarrow \quad \mathcal{B}(r) = -\frac{12\, r^6 \kappa^2}{-192\,\alpha + \kappa^2 \left(4 - 4\, r^6 - 3\,\Lambda + 3\, r^8 \Lambda\right)}.$$

$$\Rightarrow \quad R_{11}: \quad -\frac{4}{r^8} + \frac{192\,\alpha}{r^8 \kappa^2} + \Lambda + \frac{3\,\Lambda}{r^8} = \Lambda.$$

R_{11} can be solved for α:

$$\alpha = \frac{4 - 3\,\Lambda}{192} \kappa$$

$$\Rightarrow \quad \mathcal{B}(r) = \frac{12}{4 - 3\, r^2 \Lambda}.$$

We have found an exact solution to the Λ field equations. The formula for $\mathcal{D}$, above, was based upon the denominators being proportional, for which there is no requirement. Thus, there could be other, more general solutions.

a(t) is a free parameter,

$$A(t,r) = \frac{c^2 r^2 \kappa \dot{a}[t]^2}{2\zeta + \kappa a[t]^2}$$ where κ and ζ are free constants,

$$B(t,r) = \frac{12}{4 - 3 r^2 \Lambda},$$

$$\mathbb{C}(t,r) = 1,$$

$$\mathbb{D}(t,r) = a(t)^2 \mathcal{D}(r) = a(t)^2 \frac{r^2 \kappa}{8}$$

and line element

$$ds^2 = \frac{c^2 r^2 \kappa \dot{a}[t]^2}{2\zeta + \kappa a[t]^2} dt^2 - \frac{12}{4 - 3 r^2 \Lambda} dr^2 - r^2 d\theta^2 - r^2 \mathrm{Sin}[\theta]^2 d\phi^2 - \frac{1}{8} p^2 r^2 \kappa a[t]^2 d\psi^2$$

bii-II: Solve A(t,r)=c² for a(t)

There is no formula for a(t) that makes $A(t,r)=c^2$:

Solving $\frac{c^2 r^2 \kappa \dot{a}[t]^2}{2\zeta + \kappa a[t]^2} = c^2$ yields

$$a(t) = -\frac{\sqrt{2}\sqrt{\zeta}\,\mathrm{Sinh}\left[\frac{t}{r} + \sqrt{\kappa}\,\chi\right]}{\sqrt{\kappa}},$$ a function of r as well as t #.

biii: B(t,r)=B(r)=1, C(t,r)=C(r)=1

We check if there is a solution to Case bi with B(t,r)=1.

Starting fresh, using $\mathbb{B}(r)=\mathbb{C}(r)=1$, the field equations become

$$R_{00}: \quad \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{A'}{A r} + \frac{(A')^2}{4 A^2} - \frac{A' \mathcal{D}'}{4 A \mathcal{D}} - \frac{A''}{2 A} = \Lambda$$

$$R_{01}: \quad -\frac{\dot{a} A'}{2 a A} + \frac{\dot{a} \mathcal{D}'}{2 a \mathcal{D}} = 0$$

$$R_{11}: \quad \frac{(A')^2}{4 A^2} + \frac{(\mathcal{D}')^2}{4 \mathcal{D}^2} - \frac{A''}{2 A} - \frac{\mathcal{D}''}{2 \mathcal{D}} = \Lambda$$

$$R_{22}: \quad -\frac{A'}{2 A r} - \frac{\mathcal{D}'}{2 r \mathcal{D}} = \Lambda$$

$$R_{44}: \quad \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{\mathcal{D}'}{r \mathcal{D}} - \frac{A' \mathcal{D}'}{4 A \mathcal{D}} + \frac{(\mathcal{D}')^2}{4 \mathcal{D}^2} - \frac{\mathcal{D}''}{2 \mathcal{D}} = \Lambda$$

R_{01} can be simplified and solved:

$$R_{01}: \quad \frac{A'}{A} - \frac{\mathcal{D}'}{\mathcal{D}} = 0$$

$A(t,r) = \mathcal{A}[t] \times \mathcal{D}[r]$ where $\mathcal{A}(t)$ is a new unknown.

R_{01} becomes an identity and the remaining equations become

$$R_{00}: \quad \frac{c^2 \ddot{a}}{a \mathcal{A} \mathcal{D}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2 \mathcal{D}} - \frac{\mathcal{D}'}{r \mathcal{D}} - \frac{\mathcal{D}''}{2 \mathcal{D}} = \Lambda$$

$$R_{11}: \quad \frac{(\mathcal{D}')^2}{2 \mathcal{D}^2} - \frac{\mathcal{D}''}{\mathcal{D}} = \Lambda$$

$$R_{22}: \quad -\frac{\mathcal{D}'}{r \mathcal{D}} = \Lambda$$

$$R_{44}: \quad \frac{c^2 \ddot{a}}{a \mathcal{A} \mathcal{D}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2 \mathcal{D}} - \frac{\mathcal{D}'}{r \mathcal{D}} - \frac{\mathcal{D}''}{2 \mathcal{D}} = \Lambda$$

R_{44} is now redundant, the same as R_{00}.

R_{22} can be solved for $\mathcal{D}$:

$\mathcal{D}(r) = e^{-\frac{r^2 \Lambda}{2}} \alpha$ where α is a constant of integration.

Since $g_{44} = a(t)^2 \mathcal{D}(r) p^2$, we can 'fold' α into the radius p:

$$\mathcal{D}(r) = e^{-\frac{r^2 \Lambda}{2}}$$

R_{22} has become an identity, and the remaining equations become

$$R_{00}: \quad \frac{3 \Lambda}{2} - \frac{r^2 \Lambda^2}{2} + \frac{c^2 e^{\frac{r^2 \Lambda}{2}} \ddot{a}}{a \mathcal{A}} - \frac{c^2 e^{\frac{r^2 \Lambda}{2}} \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} = \Lambda$$

$$R_{11}: \quad \Lambda - \frac{r^2 \Lambda^2}{2} = \Lambda$$

R_{11} is an identity iff $\Lambda = 0$.

This simplifies $\mathcal{D}$ and A:

$$\mathcal{D}(r) = 1$$

$$A(t,r) = \mathcal{A}[t]$$

With Λ=0, the remaining equation, R_{00}, also simplifies, to

$$R_{00}: \quad \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} = 0$$

R_{00} can be solved for $\mathcal{A}$:

$\mathcal{A}(t) = k \dot{a}[t]^2$ where k is a constant of integration.

So, there is an exact solution to the Λ=0 field equations having both $\mathbb{B}(r)$=1 and $\mathbb{C}(r)$=1, but it is rather trivial.

$a(t)$ is a free parameter,

$A(t,r) = k\,\dot{a}[t]^2$ where k is a free constant,

$B(t,r) = 1$,

$\mathbb{C}(t,r) = 1$,

$\mathbb{D}(t,r) = a(t)^2 \mathbb{D}(r) = a(t)^2$,

and line element

$$ds^2 = k\,\dot{a}[t]^2\,dt^2 - 1\,dr^2 - r^2\,d\theta^2 - r^2\,\mathrm{Sin}[\theta]^2\,d\phi^2 - p^2\,a[t]^2\,d\psi^2$$

biii-II: Solve $A(t,r)=c^2$ for $a(t)$

Solving $\dfrac{k\,\dot{a}[t]^2}{c^2} = 1$ yields

$a(t) = K \pm \dfrac{c\,t}{\sqrt{k}}$ where K is a free constant.

c: $B(t,r)=B(r)$, $\mathbb{C}(t,r)=a(t)^2$

The field equations become

$$R_{00}:\quad \frac{3c^2\ddot{a}}{a\,A} - \frac{3c^2\dot{a}\dot{A}}{2a\,A^2} - \frac{A'}{A\,r\,B} + \frac{(A')^2}{4A^2B} - \frac{A'\mathbb{D}'}{4A\mathbb{D}B} + \frac{A'B'}{4AB^2} - \frac{A''}{2AB} = \Lambda$$

$$R_{01}:\quad \frac{2\dot{a}}{a\,r} - \frac{3\dot{a}A'}{2aA} + \frac{\dot{a}\mathbb{D}'}{2a\mathbb{D}} = 0$$

$$R_{11}:\quad \frac{(A')^2}{4A^2B} + \frac{(\mathbb{D}')^2}{4\mathbb{D}^2B} + \frac{B'}{r\,B^2} + \frac{A'B'}{4AB^2} + \frac{\mathbb{D}'B'}{4\mathbb{D}B^2} - \frac{A''}{2AB} - \frac{\mathbb{D}''}{2\mathbb{D}B} = \Lambda$$

$$R_{22}:\quad \frac{1}{a^2r^2} - \frac{1}{r^2B} + \frac{2c^2\dot{a}^2}{a^2A} + \frac{c^2\ddot{a}}{aA} - \frac{c^2\dot{a}\dot{A}}{2aA^2} - \frac{A'}{2Ar\,B} - \frac{\mathbb{D}'}{2r\mathbb{D}B} + \frac{B'}{2r\,B^2} = \Lambda$$

$$R_{44}:\quad \frac{2c^2\dot{a}^2}{a^2A} + \frac{c^2\ddot{a}}{aA} - \frac{c^2\dot{a}\dot{A}}{2aA^2} - \frac{\mathbb{D}'}{r\mathbb{D}B} - \frac{A'\mathbb{D}'}{4A\mathbb{D}B} + \frac{(\mathbb{D}')^2}{4\mathbb{D}^2B} + \frac{\mathbb{D}'B'}{4\mathbb{D}B^2} - \frac{\mathbb{D}''}{2\mathbb{D}B} = \Lambda$$

Since $a(t)$ is not allowed to be constant in Case 1A, $\dfrac{\dot{a}}{a}$ can be factored out, simplifying R_{01}, which can then be solved:

$$R_{01}:\quad \frac{2}{r} - \frac{3A'}{2A} + \frac{\mathbb{D}'}{2\mathbb{D}} = 0$$

$A[t,r] = r^{4/3}\,\mathcal{A}[t]\,\mathbb{D}[r]^{1/3}$ where $\mathcal{A}(t)$ is a new unknown.

This eliminates R_{01}, and the remaining equations become

$$R_{00}:\quad -\frac{10}{9r^2B} + \frac{3c^2\ddot{a}}{a\,r^{4/3}\mathcal{A}\mathbb{D}^{1/3}} - \frac{3c^2\dot{a}\dot{\mathcal{A}}}{2a\,r^{4/3}\mathcal{A}^2\mathbb{D}^{1/3}} - \frac{8\mathbb{D}'}{9r\mathbb{D}B} + \frac{(\mathbb{D}')^2}{18\mathbb{D}^2B} + \frac{B'}{3r\,B^2} + \frac{\mathbb{D}'B'}{12\mathbb{D}B^2} - \frac{\mathbb{D}''}{6\mathbb{D}B} = \Lambda$$

$$R_{11}: \quad \frac{2}{9\,r^2\,\mathbb{B}} - \frac{2\,\mathbb{D}'}{9\,r\,\mathbb{D}\,\mathbb{B}} + \frac{7\,(\mathbb{D}')^2}{18\,\mathbb{D}^2\,\mathbb{B}} + \frac{4\,\mathbb{B}'}{3\,r\,\mathbb{B}^2} + \frac{\mathbb{D}'\,\mathbb{B}'}{3\,\mathbb{D}\,\mathbb{B}^2} - \frac{2\,\mathbb{D}''}{3\,\mathbb{D}\,\mathbb{B}} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2} - \frac{5}{3\,r^2\,\mathbb{B}} + \frac{2\,c^2\,\dot{a}^2}{a^2\,r^{4/3}\,\mathcal{A}\,\mathbb{D}^{1/3}} + \frac{c^2\,\ddot{a}}{a\,r^{4/3}\,\mathcal{A}\,\mathbb{D}^{1/3}} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,r^{4/3}\,\mathcal{A}^2\,\mathbb{D}^{1/3}} - \frac{2\,\mathbb{D}'}{3\,r\,\mathbb{D}\,\mathbb{B}} + \frac{\mathbb{B}'}{2\,r\,\mathbb{B}^2} = \Lambda$$

$$R_{44}: \quad \frac{2\,c^2\,\dot{a}^2}{a^2\,r^{4/3}\,\mathcal{A}\,\mathbb{D}^{1/3}} + \frac{c^2\,\ddot{a}}{a\,r^{4/3}\,\mathcal{A}\,\mathbb{D}^{1/3}} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,r^{4/3}\,\mathcal{A}^2\,\mathbb{D}^{1/3}} - \frac{4\,\mathbb{D}'}{3\,r\,\mathbb{D}\,\mathbb{B}} + \frac{(\mathbb{D}')^2}{6\,\mathbb{D}^2\,\mathbb{B}} + \frac{\mathbb{D}'\,\mathbb{B}'}{4\,\mathbb{D}\,\mathbb{B}^2} - \frac{\mathbb{D}''}{2\,\mathbb{D}\,\mathbb{B}} = \Lambda$$

I found that I could algebraically solve $4R_{00}-R_{11}$ for $\mathbb{B}$, algebraically solve either $R_{00}-3R_{22}$, $R_{00}-3R_{44}$, or $4R_{00}-R_{11}-12R_{44}$ for $\mathcal{A}$, Dsolve R_{11} or $R_{22}-R_{44}$ for $\mathbb{B}$, or DSolve $3R_{11}+4R_{22}-4R_{44}$ for either $\mathbb{B}$ or $\mathbb{D}$, but all these attempts failed to generate useful equations.

What works is to solve $3R_{00}-2R_{22}-R_{44}$ for $\mathcal{A}$:

$$3R_{00}-2R_{22}-R_{44}: \quad -\frac{2}{a^2\,r^2} - \frac{6\,c^2\,\dot{a}^2}{a^2\,r^{4/3}\,\mathcal{A}\,\mathbb{D}^{1/3}} + \frac{6\,c^2\,\ddot{a}}{a\,r^{4/3}\,\mathcal{A}\,\mathbb{D}^{1/3}} - \frac{3\,c^2\,\dot{a}\,\dot{\mathcal{A}}}{a\,r^{4/3}\,\mathcal{A}^2\,\mathbb{D}^{1/3}} = 0$$

$$\mathcal{A}(t) = \frac{3\,c^2\,r^{2/3}\,\dot{a}^2}{3\,a^2\,c^2\,k\,r^{2/3} - \mathbb{D}^{1/3}}$$

In order for $\mathcal{A}(t)$ to be strictly a function of t, $\mathbb{D}^{1/3} \propto r^{2/3}$; that is

$\mathbb{D}(r) = \mathbb{K}^3\,r^2$ for some proportionality constant $\mathbb{K}$.

Using this formula, $\mathcal{A}(t)$ clearly becomes a strict function of t:

$$\mathcal{A}(t) = -\frac{3\,c^2\,\dot{a}[t]^2}{\mathbb{K} - 3\,c^2\,k\,a[t]^2}$$

The equations now have only one unknown, $\mathbb{B}$:

$$R_{00}: \quad -\frac{3}{r^2\,\mathbb{B}} + \frac{3\,c^2\,k\,\left(r^2\,\mathbb{K}^3\right)^{2/3}}{r^{10/3}\,\mathbb{K}^3} + \frac{\mathbb{B}'}{2\,r\,\mathbb{B}^2} = \Lambda$$

$$R_{11}: \quad \frac{2\,\mathbb{B}'}{r\,\mathbb{B}^2} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2} - \frac{3}{r^2\,\mathbb{B}} + \frac{3\,c^2\,k\,\left(r^2\,\mathbb{K}^3\right)^{2/3}}{r^{10/3}\,\mathbb{K}^3} - \frac{2\,\left(r^2\,\mathbb{K}^3\right)^{2/3}}{3\,a^2\,r^{10/3}\,\mathbb{K}^2} + \frac{\mathbb{B}'}{2\,r\,\mathbb{B}^2} = \Lambda$$

$$R_{44}: \quad -\frac{3}{r^2\,\mathbb{B}} + \frac{3\,c^2\,k\,\left(r^2\,\mathbb{K}^3\right)^{2/3}}{r^{10/3}\,\mathbb{K}^3} - \frac{2\,\left(r^2\,\mathbb{K}^3\right)^{2/3}}{3\,a^2\,r^{10/3}\,\mathbb{K}^2} + \frac{\mathbb{B}'}{2\,r\,\mathbb{B}^2} = \Lambda$$

Solving R_{11} yields

$$\mathbb{B}(r) = -\frac{4}{4\,\beta + r^2\,\Lambda}.$$

Plugging this in, we get

R_{00}: $\frac{3\beta}{r^2} + \frac{3 c^2 k \left(r^2 K^3\right)^{2/3}}{r^{10/3} K^3} + \Lambda = \Lambda$

R_{22}: $\frac{1}{a^2 r^2} + \frac{3\beta}{r^2} + \frac{3 c^2 k \left(r^2 K^3\right)^{2/3}}{r^{10/3} K^3} - \frac{2 \left(r^2 K^3\right)^{2/3}}{3 a^2 r^{10/3} K^2} + \Lambda = \Lambda$

R_{44}: $\frac{3\beta}{r^2} + \frac{3 c^2 k \left(r^2 K^3\right)^{2/3}}{r^{10/3} K^3} - \frac{2 \left(r^2 K^3\right)^{2/3}}{3 a^2 r^{10/3} K^2} + \Lambda = \Lambda$

and $R_{22}-R_{44}$: $\frac{1}{r^2 a[t]^2} = 0$, which has no solution.

This case has no possible solution.

We now re-do this case with the additional assumption that $A(t,r) = A(r)$, which is a **subcase, Iβ**, of Case I: $A(t,r)=\mathcal{A}(t)A(r)$. This assumption changes R_{01} and could lead to a solution to the equations.

c_Iα: $B(t,r)=B(r)$, $\mathbb{C}(t,r)=a(t)^2$, $A(t,r)=\mathcal{A}(t)$

We attempt to solve Case c again but with the extra condition Iα, that $A(t,r)=\mathcal{A}(t)$. Starting fresh, the field equations are identical to those in Case c except that all instances of $A(t,r)$ are replaced by $\mathcal{A}(t)$:

R_{00}: $\frac{3 c^2 \ddot{a}}{a \mathcal{A}} - \frac{3 c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} = \Lambda$

R_{01}: $\frac{2 \dot{a}}{a r} + \frac{\dot{a} \mathcal{D}'}{2 a \mathcal{D}} = 0$

R_{11}: $\frac{(\mathcal{D}')^2}{4 \mathcal{D}^2 B} + \frac{B'}{r B^2} + \frac{\mathcal{D}' B'}{4 \mathcal{D} B^2} - \frac{\mathcal{D}''}{2 \mathcal{D} B} = \Lambda$

R_{22}: $\frac{1}{a^2 r^2} - \frac{1}{r^2 B} + \frac{2 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{\mathcal{D}'}{2 r \mathcal{D} B} + \frac{B'}{2 r B^2} = \Lambda$

R_{44}: $\frac{2 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{\mathcal{D}'}{r \mathcal{D} B} + \frac{(\mathcal{D}')^2}{4 \mathcal{D}^2 B} + \frac{\mathcal{D}' B'}{4 \mathcal{D} B^2} - \frac{\mathcal{D}''}{2 \mathcal{D} B} = \Lambda$

We can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then solving for $\mathcal{D}$:

R_{01}: $-\frac{2}{r} - \frac{\mathcal{D}'}{2 \mathcal{D}} = 0$

$\mathcal{D}[r] = \frac{k}{r^4}$ where k is a constant of integration

This eliminates R_{01}, and the remaining 4 equations become

R_{00}: $\frac{3 c^2 \ddot{a}}{a \mathcal{A}} - \frac{3 c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} = \Lambda$

$$R_{11}: \quad -\frac{6}{r^2 B} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2} + \frac{1}{r^2 B} + \frac{2 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} + \frac{B'}{2 r B^2} = \Lambda$$

$$R_{44}: \quad -\frac{2}{r^2 B} + \frac{2 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{B'}{r B^2} = \Lambda$$

$R_{22}-R_{44}$ has no solution:

$$R_{22}-R_{44}: \quad \frac{1}{r^2 a[t]^2} + \frac{3}{r^2 B[r]} + \frac{3 B'[r]}{2 r B[r]^2} = 0$$

This case has no solution.

c_Iβ: B(t,r)=B(r), C(t,r)=a(t)^2, A(t,r)=A(r)

We attempt to solve Case c again but with the extra condition Iβ, that A(t,r)=A(r). Starting fresh, the field equations are identical to those in Case c except that all instances of A(t,r) are replaced by A(r):

$$R_{00}: \quad \frac{3 c^2 \ddot{a}}{a A} - \frac{A'}{r A B} - \frac{D' A'}{4 D A B} + \frac{(A')^2}{4 A^2 B} + \frac{A' B'}{4 A B^2} - \frac{A''}{2 A B} = \Lambda$$

$$R_{01}: \quad \frac{2 \dot{a}}{a r} + \frac{\dot{a} D'}{2 a D} - \frac{3 \dot{a} A'}{2 a A} = 0$$

$$R_{11}: \quad \frac{(D')^2}{4 D^2 B} + \frac{(A')^2}{4 A^2 B} + \frac{B'}{r B^2} + \frac{D' B'}{4 D B^2} + \frac{A' B'}{4 A B^2} - \frac{D''}{2 D B} - \frac{A''}{2 A B} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2} - \frac{1}{r^2 B} + \frac{2 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{D'}{2 r D B} - \frac{A'}{2 r A B} + \frac{B'}{2 r B^2} = \Lambda$$

$$R_{44}: \quad \frac{2 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{D'}{r D B} + \frac{(D')^2}{4 D^2 B} - \frac{D' A'}{4 D A B} + \frac{D' B'}{4 D B^2} - \frac{D''}{2 D B} = \Lambda$$

We can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then solving for D:

$$R_{01}: \quad -\frac{2}{r} - \frac{D'}{2 D} + \frac{3 A'}{2 A} = 0$$

$$D[r] = \frac{k A^3}{r^4} \quad \text{where k is a constant of integration}$$

This eliminates R_{01}, and the remaining 4 equations become

$$R_{00}: \quad \frac{3 c^2 \ddot{a}}{a A} - \frac{(A')^2}{2 A^2 B} + \frac{A' B'}{4 A B^2} - \frac{A''}{2 A B} = \Lambda$$

$$R_{11}: \quad -\frac{6}{r^2 B} + \frac{6 A'}{r A B} - \frac{(A')^2}{2 A^2 B} + \frac{A' B'}{A B^2} - \frac{2 A''}{A B} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2} + \frac{1}{r^2 B} + \frac{2 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{2 A'}{r A B} + \frac{B'}{2 r B^2} = \Lambda$$

$$R_{44}: \quad -\frac{2}{r^2 B} + \frac{2 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} + \frac{4 A'}{r A B} - \frac{3 (A')^2}{2 A^2 B} - \frac{B'}{r B^2} + \frac{3 A' B'}{4 A B^2} - \frac{3 A''}{2 A B} = \Lambda$$

Now we can algebraically solve for A:

$$3R_{00}-2R_{22}-R_{44}: \quad -\frac{2}{a^2 r^2} - \frac{6 c^2 \dot{a}^2}{a^2 A} + \frac{6 c^2 \ddot{a}}{a A} = 0$$

$$\Rightarrow \quad A(r) = -3 c^2 r^2 \left(\dot{a}[t]^2 - a[t]\, \ddot{a}[t]\right)$$

In order for $A(r)$ to be strictly a function of r,
the expression in the parentheses must equal a constant, say K:
Solving for this yields

$$a(t) = -\frac{\sqrt{K}\, \mathrm{Sinh}[t\,\alpha + \beta]}{\alpha} \quad \text{where } \alpha,\ \beta, \text{ and K are constants}$$

$$\Rightarrow \quad A(r) = -3 c^2 K r^2$$

With this expression for A, (but ignoring for now that $a(t)$ has a formula), we have $R_{44}=3R_{00}-2R_{22}$, which makes R_{44} redundant, and the field equations become:

$$R_{00}: \quad -\frac{3}{r^2 B} - \frac{\ddot{a}}{a K r^2} + \frac{B'}{2 r B^2} = \Lambda$$

$$R_{11}: \quad \frac{2 B'}{r B^2} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2} - \frac{3}{r^2 B} - \frac{2 \dot{a}^2}{3 a^2 K r^2} - \frac{\ddot{a}}{3 a K r^2} + \frac{B'}{2 r B^2} = \Lambda$$

$R_{00}-R_{22}$ has no solution:

$$R_{00}-R_{22}: \quad -\frac{1}{r^2 a[t]^2} + \frac{2 \dot{a}[t]^2}{3 K r^2 a[t]^2} - \frac{2 \ddot{a}[t]}{3 K r^2 a[t]} = 0$$

Plugging in $a(t)$ makes this

$$R_{00}-R_{22}: \quad -\frac{\alpha^2 \mathrm{Csch}[t\,\alpha + \beta]^2}{3 K r^2} = 0$$

Since $\mathrm{Csch}^2(t)>0$ for all t, we must have

$$\alpha = 0$$

which makes $a(t)$ undefined.

This case has no possible solution.

c_II: $B(t,r)=B(r)$, $C(t,r)=a(t)^2$, $A(t,r)=c^2$

We try to solve Case c again, but with the stronger extra condition that $A(t,r)=c^2$.

Starting again from scratch, the field equations become

$$R_{00}: \quad \frac{3\,\ddot{a}}{a} = \Lambda$$

$$R_{01}: \quad \frac{2\,\dot{a}}{a\,r} + \frac{\dot{a}\,D'}{2\,a\,D} = 0$$

$$R_{11}: \quad \frac{(D')^2}{4\,D^2\,B} + \frac{B'}{r\,B^2} + \frac{D'\,B'}{4\,D\,B^2} - \frac{D''}{2\,D\,B} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2} - \frac{1}{r^2\,B} + \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{D'}{2\,r\,D\,B} + \frac{B'}{2\,r\,B^2} = \Lambda$$

$$R_{44}: \quad \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{D'}{r\,D\,B} + \frac{(D')^2}{4\,D^2\,B} + \frac{D'\,B'}{4\,D\,B^2} - \frac{D''}{2\,D\,B} = \Lambda$$

Solving R_{00} yields

$$a(t) = e^{\frac{t\sqrt{\Lambda}}{\sqrt{3}}}\,\alpha + e^{-\frac{t\sqrt{\Lambda}}{\sqrt{3}}}\,\beta$$ where α and β are constants of integration.

It is easier for now to work with $a(t)$ rather than its fomula, noting just that R_{00} is an identity and can be ignored.

R_{01} can be simplified and then solved for D:

$$R_{01}: \quad -\frac{4}{r} - \frac{D'}{D} = 0$$

$$D(r) = \frac{k}{r^4}$$ where k is a constant of integration.

R_{01} is now also an identity. Plugging in $D(r)$, the remaining equations become

$$R_{11}: \quad -\frac{6}{r^2\,B} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2} + \frac{1}{r^2\,B} + \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{B'}{2\,r\,B^2} = \Lambda$$

$$R_{44}: \quad -\frac{2}{r^2\,B} + \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{B'}{r\,B^2} = \Lambda$$

From R_{11} we see that

$$B(r) = -\frac{6}{\Lambda\,r^2}.$$

Plugging this in, we have

$$R_{22}: \quad \frac{1}{a^2\,r^2} + \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} = \Lambda$$

$$R_{44}: \quad \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} = \Lambda$$

and $R_{22}-R_{44}$: $\dfrac{1}{r^2\,a[t]^2} = 0$, which is false.

This contradiction shows that this case has no solution.

d: $B(t,r)=a(t)^2$, $\mathbb{C}(t,r)=\mathbb{C}(r)$

The field equations become

$$R_{00}: \quad \frac{2\,c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{a\,A^2} - \frac{A'}{a^2\,A\,r} + \frac{(A')^2}{4\,a^2\,A^2} - \frac{A'\,\mathbb{C}'}{2\,a^2\,A\,\mathbb{C}} - \frac{A'\,\mathbb{D}'}{4\,a^2\,A\,\mathbb{D}} - \frac{A''}{2\,a^2\,A} = \Lambda$$

$$R_{01}: \quad -\frac{2\,\dot{a}}{a\,r} - \frac{\dot{a}\,A'}{2\,a\,A} - \frac{\dot{a}\,\mathbb{C}'}{a\,\mathbb{C}} = 0$$

$$R_{11}: \quad \frac{c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} + \frac{(A')^2}{4\,a^2\,A^2} - \frac{2\,\mathbb{C}'}{a^2\,r\,\mathbb{C}} + \frac{(\mathbb{C}')^2}{2\,a^2\,\mathbb{C}^2} + \frac{(\mathbb{D}')^2}{4\,a^2\,\mathbb{D}^2} - \frac{A''}{2\,a^2\,A} - \frac{\mathbb{C}''}{a^2\,\mathbb{C}} - \frac{\mathbb{D}''}{2\,a^2\,\mathbb{D}} = \Lambda$$

$$R_{22}: \quad -\frac{1}{a^2\,r^2} + \frac{1}{r^2\,\mathbb{C}} - \frac{A'}{2\,a^2\,A\,r} - \frac{2\,\mathbb{C}'}{a^2\,r\,\mathbb{C}} - \frac{A'\,\mathbb{C}'}{4\,a^2\,A\,\mathbb{C}} - \frac{\mathbb{D}'}{2\,a^2\,r\,\mathbb{D}} - \frac{\mathbb{C}'\,\mathbb{D}'}{4\,a^2\,\mathbb{C}\,\mathbb{D}} - \frac{\mathbb{C}''}{2\,a^2\,\mathbb{C}} = \Lambda$$

$$R_{44}: \quad \frac{c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{\mathbb{D}'}{a^2\,r\,\mathbb{D}} - \frac{A'\,\mathbb{D}'}{4\,a^2\,A\,\mathbb{D}} - \frac{\mathbb{C}'\,\mathbb{D}'}{2\,a^2\,\mathbb{C}\,\mathbb{D}} + \frac{(\mathbb{D}')^2}{4\,a^2\,\mathbb{D}^2} - \frac{\mathbb{D}''}{2\,a^2\,\mathbb{D}} = \Lambda$$

Because $a(t)\neq$constant in Case 1A, $\dot{a}\neq 0$,
allowing R_{01} to be simplified. It can then be solved for A:

$$R_{01}: \quad -\frac{2}{r} - \frac{A'}{2\,A} - \frac{\mathbb{C}'}{\mathbb{C}} = 0$$

$A(t,r) = \dfrac{\mathcal{A}}{r^4\,\mathbb{C}^2}$ where $\mathcal{A}(t)$ is a new unknown.

This eliminates R_{01}, and the remaining equations become

$$R_{00}: \quad -\frac{2}{a^2\,r^2} + \frac{2\,c^2\,r^4\,\mathbb{C}^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{c^2\,r^4\,\mathbb{C}^2\,\dot{a}\,\dot{\mathcal{A}}}{a\,\mathcal{A}^2} - \frac{(\mathbb{C}')^2}{a^2\,\mathbb{C}^2} + \frac{\mathbb{D}'}{a^2\,r\,\mathbb{D}} + \frac{\mathbb{C}'\,\mathbb{D}'}{2\,a^2\,\mathbb{C}\,\mathbb{D}} + \frac{\mathbb{C}''}{a^2\,\mathbb{C}} = \Lambda$$

$$R_{11}: \quad -\frac{6}{a^2\,r^2} + \frac{c^2\,r^4\,\mathbb{C}^2\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{c^2\,r^4\,\mathbb{C}^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{c^2\,r^4\,\mathbb{C}^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} - \frac{6\,\mathbb{C}'}{a^2\,r\,\mathbb{C}} - \frac{3\,(\mathbb{C}')^2}{2\,a^2\,\mathbb{C}^2} + \frac{(\mathbb{D}')^2}{4\,a^2\,\mathbb{D}^2} - \frac{\mathbb{D}''}{2\,a^2\,\mathbb{D}} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2} + \frac{1}{r^2\,\mathbb{C}} + \frac{(\mathbb{C}')^2}{2\,a^2\,\mathbb{C}^2} - \frac{\mathbb{D}'}{2\,a^2\,r\,\mathbb{D}} - \frac{\mathbb{C}'\,\mathbb{D}'}{4\,a^2\,\mathbb{C}\,\mathbb{D}} - \frac{\mathbb{C}''}{2\,a^2\,\mathbb{C}} = \Lambda$$

$$R_{44}: \quad \frac{c^2\,r^4\,\mathbb{C}^2\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{c^2\,r^4\,\mathbb{C}^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{c^2\,r^4\,\mathbb{C}^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} + \frac{(\mathbb{D}')^2}{4\,a^2\,\mathbb{D}^2} - \frac{\mathbb{D}''}{2\,a^2\,\mathbb{D}} = \Lambda$$

We can solve $R_{11}-R_{44}$ for $\mathbb{C}$:

$$R_{11}-R_{44}: \quad -\frac{6}{a^2\,r^2} - \frac{6\,\mathbb{C}'}{a^2\,r\,\mathbb{C}} - \frac{3\,(\mathbb{C}')^2}{2\,a^2\,\mathbb{C}^2} = 0$$

$\Rightarrow \quad \mathbb{C}(r) = \frac{K}{r^2}$ where K is a constant of integration.

$\Rightarrow \quad A(t,r) = \frac{\mathcal{A}[t]}{K^2}$.

Plugging $\mathbb{C}(r)$ back in yields four equations in the unknowns $\mathcal{A}$ and $\mathbb{D}$:

$$R_{00}: \quad \frac{2\,c^2\,K^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{c^2\,K^2\,\dot{a}\,\dot{\mathcal{A}}}{a\,\mathcal{A}^2} = \Lambda$$

$$R_{11}: \quad \frac{c^2\,K^2\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{c^2\,K^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{c^2\,K^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} + \frac{(\mathbb{D}')^2}{4\,a^2\,\mathbb{D}^2} - \frac{\mathbb{D}''}{2\,a^2\,\mathbb{D}} = \Lambda$$

$$R_{22}: \quad \frac{1}{K} = \Lambda$$

$$R_{44}: \quad \frac{c^2\,K^2\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{c^2\,K^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{c^2\,K^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} + \frac{(\mathbb{D}')^2}{4\,a^2\,\mathbb{D}^2} - \frac{\mathbb{D}''}{2\,a^2\,\mathbb{D}} = \Lambda$$

(1) R_{44} is now redundant, the same as R_{11},

(2) from R_{22} we have

$\Lambda \neq 0$,

$K = \frac{1}{\Lambda}$,

$\Rightarrow \quad A(t,r) = \Lambda^2\,\mathcal{A}[t]$,

$\Rightarrow \quad \mathbb{C}(r) = \frac{1}{r^2\,\Lambda}$,

and

(3) R_{00} can be solved to get

$\mathcal{A}(t) = \frac{2\,c^2\,\dot{a}[t]^2}{2\,\zeta^2\,\Lambda^2 + \Lambda^3\,a[t]^2}$ where ζ is a constant of integration.

$\Rightarrow \quad A(t,r) = \frac{2\,c^2\,\dot{a}[t]^2}{2\,\zeta^2 + \Lambda\,a[t]^2}$

Applying this information, the one remaining equation becomes

$$R_{11}: \quad \frac{\zeta^2}{a^2} + \Lambda + \frac{(\mathbb{D}')^2}{4\,a^2\,\mathbb{D}^2} - \frac{\mathbb{D}''}{2\,a^2\,\mathbb{D}} = \Lambda$$

Solving this equations yields

$\mathbb{D}(r) = X\,\mathrm{Cosh}[\zeta\,(r-\omega)]^2$ where X and ω are constants of integration.

Since $g_{44} = -a(t)^2\,\mathbb{D}(r)\,p^2$, X can be absorbed into the radius p:

$\mathbb{D}(r) = \mathrm{Cosh}[\zeta\,(r-\omega)]^2$.

All of the equations have been satisfied,
so we have an exact solution to the $\Lambda\neq 0$ field equations:

$a(t)$ is a free parameter,

$$A(t,r) = \Lambda^2\,\mathcal{A}(t) = \frac{2\,c^2\,\dot{a}[t]^2}{2\,\zeta^2+\Lambda\,a[t]^2} \quad \text{where } \zeta \text{ is a free constant,}$$

$$B(t,r) = a[t]^2,$$

$$\mathbb{C}(t,r) = \mathbb{C}(r) = \frac{1}{r^2\,\Lambda} \quad \text{where } \Lambda\neq 0,$$

$$\mathbb{D}(t,r) = a(t)^2\,\mathbb{D}(r) = a[t]^2\,\mathrm{Cosh}[\zeta\,(r-\omega)]^2$$

where ω is a free constant,

and line element

$$ds^2 = \frac{2\,c^2\,\dot{a}[t]^2}{2\,\zeta^2+\Lambda\,a[t]^2}\,dt^2 - a[t]^2\,dr^2 - \frac{1}{\Lambda}\,d\theta^2 - \frac{\mathrm{Sin}[\theta]^2}{\Lambda}\,d\phi^2 - p^2\,a[t]^2\,\mathrm{Cosh}[\zeta\,(r-\omega)]^2\,d\psi^2$$

d-II: Solve $A(t,r)=c^2$ for $a(t)$

Solving $\dfrac{2\,c^2\,\dot{a}[t]^2}{2\,\zeta^2\,\Lambda^2+\Lambda^3\,a[t]^2} = c^2$ yields

$$a(t) = \frac{\sqrt{2}\,\zeta\,\mathrm{Sinh}\left[\sqrt{\Lambda}\,\left(\pm\frac{t}{\sqrt{2}}+\chi\right)\right]}{\sqrt{\Lambda}} \quad \text{where } \chi \text{ is a constant of integration.}$$

d_II: $B(t,r)=a(t)^2$, $\mathbb{C}(t,r)=\mathbb{C}(r)$, $A(t,r)=c^2$

We solve Case d again, but with the extra condition that $A(t,r)=c^2$. Even though a solution to this case can be achieved by plugging $A(t,r)=c^2$ and the formula for $a(t)$, above, into the Case d solution, it is possible that a different solution can be obtained by assuming $A(t,r)=c^2$ before the ODEs are solved, rather than after.

Starting again from scratch, the field equations become

$$R_{00}:\quad \frac{2\,\ddot{a}}{a} = \Lambda$$

$$R_{01}:\quad -\frac{2\,\dot{a}}{a\,r} - \frac{\dot{a}\,\mathbb{C}'}{a\,\mathbb{C}} = 0$$

$$R_{11}:\quad \frac{\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{2\,\mathbb{C}'}{a^2\,r\,\mathbb{C}} + \frac{(\mathbb{C}')^2}{2\,a^2\,\mathbb{C}^2} + \frac{(\mathbb{D}')^2}{4\,a^2\,\mathbb{D}^2} - \frac{\mathbb{C}''}{a^2\,\mathbb{C}} - \frac{\mathbb{D}''}{2\,a^2\,\mathbb{D}} = \Lambda$$

$$R_{22}:\quad -\frac{1}{a^2\,r^2} + \frac{1}{r^2\,\mathbb{C}} - \frac{2\,\mathbb{C}'}{a^2\,r\,\mathbb{C}} - \frac{\mathbb{D}'}{2\,a^2\,r\,\mathbb{D}} - \frac{\mathbb{C}'\,\mathbb{D}'}{4\,a^2\,\mathbb{C}\,\mathbb{D}} - \frac{\mathbb{C}''}{2\,a^2\,\mathbb{C}} = \Lambda$$

$$R_{44}:\quad \frac{\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{\mathbb{D}'}{a^2\,r\,\mathbb{D}} - \frac{\mathbb{C}'\,\mathbb{D}'}{2\,a^2\,\mathbb{C}\,\mathbb{D}} + \frac{(\mathbb{D}')^2}{4\,a^2\,\mathbb{D}^2} - \frac{\mathbb{D}''}{2\,a^2\,\mathbb{D}} = \Lambda$$

Solving R_{00} yields

$$a(t) = e^{\frac{t\sqrt{\Lambda}}{\sqrt{2}}}\,\alpha + e^{-\frac{t\sqrt{\Lambda}}{\sqrt{2}}}\,\beta \quad \text{where } \alpha \text{ and } \beta \text{ are constants of integration.}$$

This formula for a(t) will be reflected in the solution, but for now it is easier just to work with the symbol, a(t), noting that R_{00} has become an identity and can thus be ignored.

R_{01} can be simplified and then solved for $\mathbb{D}$:

$$R_{01}: \quad \frac{4}{r} + \frac{2\,\mathbb{C}'}{\mathbb{C}} = 0$$

$$\mathbb{C}(r) = \frac{\xi}{r^2}$$ where ξ is a constant of integration.

R_{01} is now also an identity. Plugging in $\mathbb{C}(r)$, the remaining equations become

$$R_{11}: \quad \frac{\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{(\mathbb{D}')^2}{4\,a^2\,\mathbb{D}^2} - \frac{\mathbb{D}''}{2\,a^2\,\mathbb{D}} = \Lambda$$

$$R_{22}: \quad \frac{1}{\xi} = \Lambda$$

$$R_{44}: \quad \frac{\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{(\mathbb{D}')^2}{4\,a^2\,\mathbb{D}^2} - \frac{\mathbb{D}''}{2\,a^2\,\mathbb{D}} = \Lambda$$

R_{44} is redundant, the same as R_{11}, and from R_{22},

$$\Lambda \neq 0$$

and

$$\xi = \frac{1}{\Lambda}$$

Solving R_{11} for $\mathbb{D}$ yields

$$\mathbb{D}(r) = \gamma\,\mathrm{Cos}\left[(r-\omega)\sqrt{\Lambda\,a[t]^2 - \dot{a}[t]^2 - a[t]\,\ddot{a}[t]}\,\right]^2$$

where γ and ω are constants of integration.

As with Case d, γ can be absorbed into the radius p:

$$\mathbb{D}(r) = \mathrm{Cos}\left[(r-\omega)\sqrt{\Lambda\,a[t]^2 - \dot{a}[t]^2 - a[t]\,\ddot{a}[t]}\,\right]^2.$$

Plugging in the formula for a(t) simplfies $\mathbb{D}$:

$$\mathbb{D}(r) = \mathrm{Cos}\left[\sqrt{2}\,\sqrt{\alpha\,\beta\,\Lambda}\,(r-\omega)\right]^2$$

All of the equations are now satisifed, and so we have obtained a solution:

$$a(t) = e^{\frac{t\sqrt{\Lambda}}{\sqrt{2}}}\,\alpha + e^{-\frac{t\sqrt{\Lambda}}{\sqrt{2}}}\,\beta$$ where α and β are free constants

$$A(t,r) = c^2,$$

$B(t,r) = a[t]^2,$

$\mathbb{C}(t,r) = \mathbb{C}(r) = \dfrac{1}{r^2\Lambda}$ where $\Lambda \neq 0$,

$\mathbb{D}(t,r) = a(t)^2\mathbb{D}(r) = a[t]^2\mathrm{Cos}\left[\sqrt{2}\sqrt{\alpha\beta\Lambda}\,(r-\omega)\right]^2$

where ω is a free constant,

and line element

$$ds^2 = c^2\,dt^2 - a[t]^2\,dr^2 - \frac{1}{\Lambda}\,d\theta^2 - \frac{\mathrm{Sin}[\theta]^2}{\Lambda}\,d\phi^2 - p^2\,a[t]^2\,\mathrm{Cos}\left[\sqrt{2}\sqrt{\alpha\beta\Lambda}\,(r-\omega)\right]^2\,d\psi^2$$

We note that this is a new solution, but it is almost a special case of the more general Case d where $A(t,r) = c^2$, and, in common practice, it is a special case.

- $A(t,r)$, $B(t,r)$ and $\mathbb{C}(t,r)$: Both solutions use the same formulas ✓
- $\mathbb{D}(t,r)$

  Set $\zeta = \sqrt{-2\alpha\beta\Lambda}$

  Then, in Case d,

  $\mathbb{D}(t,r) = a[t]^2\mathrm{Cosh}[\zeta(r-\omega)]^2 = a[t]^2\mathrm{Cosh}[i\sqrt{2\alpha\beta\Lambda}\,(r-\omega)]^2$

  $= a[t]^2\mathrm{Cos}[\sqrt{2\alpha\beta\Lambda}\,(r-\omega)]^2$

  $= \mathbb{D}(t,r)$ in Case d_II ✓
- $a(t)$

  Commonly in Case d_II we would have $\alpha = -\beta = \dfrac{1}{2}$ so that

  $$a(t) = e^{\frac{t\sqrt{\Lambda}}{\sqrt{2}}}\alpha + e^{-\frac{t\sqrt{\Lambda}}{\sqrt{2}}}\beta = -\frac{1}{2}e^{-\frac{t\sqrt{\Lambda}}{\sqrt{2}}} + \frac{1}{2}e^{\frac{t\sqrt{\Lambda}}{\sqrt{2}}} = \mathrm{Sinh}\left[\frac{t\sqrt{\Lambda}}{\sqrt{2}}\right],$$

  which is the special case of Case d with $\zeta = \sqrt{\dfrac{\Lambda}{2}}$ and $\chi = 0$ ✓